

High-Frequency Traders in the Options Market: Cream Skimming and Toxic Order Flow

Mahendrarajah Nimalendran, Matthew G. Son*

Abstract

We examine how professional customers (PCs), high-frequency options traders, influence options market microstructure. Extending Foucault et al. (2017), we develop a model where PCs' speed advantage creates toxic arbitrage (exploiting stale quotes) and non-toxic arbitrage (cream-skimming uninformed flow). Both mechanisms widen spreads, with cream-skimming dominating in low-volatility states. Using CBOE data, we find PCs engage in cream-skimming by targeting large, uninformed customer orders while avoiding informed flow and withdrawing around earnings. This behavior shifts adverse selection risk to market makers, increasing spreads by 50 basis points.

*Mahendrarajah Nimalendran is at the University of Florida (mahen.nimalendran@warrington.ufl.edu). Matthew G. Son is at the University of South Florida (email: gson@usf.edu). We thank comments from Giovanni Petrella, Jay Ritter, Yuehua Tang, Baolian Wang, Yanbin Wu, Paul Irvine, Swaminathan Kalpathy, Vassil Mihov, Doina Chichernea and other participants at conferences and seminars for valuable comments and suggestions. We acknowledge the support of high-performance cluster computing from UFIT Research Computing Lab (HiPerGator) and USF Research Computing (CIRCE).

High-frequency trading (HFT) has transformed financial markets, prompting extensive research into its impact on market quality. While prior studies have predominantly focused on HFT in equity markets, its influence on the options market remains less studied. This gap is particularly relevant given the structural differences between the two markets, including wider spreads, quote-driven dynamics, and exclusive exchange trading in options. Combined with the surge in retail participation since 2020, these distinctions raise important questions about how HFT strategies adapt to the options market and their subsequent effects on market quality.

Existing literature on HFT in equity markets presents a mixed picture. Some studies suggest positive externalities, such as improved liquidity and price efficiency (e.g., [Hendershott et al. \(2011\)](#); [Carrion \(2013\)](#); [Hendershott and Riordan \(2013\)](#); [Jones \(2013\)](#); [Menkveld \(2013\)](#); [Brogaard et al. \(2014\)](#); [O'Hara \(2015\)](#)). While other studies raise concerns about negative impacts, including increased transaction costs for slower traders and adverse selection risks for market makers (e.g., [Hoffmann \(2014\)](#); [Yao and Ye \(2018\)](#); [Li et al. \(2021\)](#)). Moreover, HFTs' ability to anticipate and front-run large institutional orders can inflate transaction costs for these institutions (e.g., [Brunnermeier and Pedersen \(2005\)](#); [Korajczyk and Murphy \(2019\)](#); [Hirschey \(2021\)](#)). HFTs' speed advantage in arbitrage activities also exposes dealers to the risk of trading at stale quotes, generating a toxic order flow for both stock markets (e.g., [Foucault et al. \(2017\)](#)) and options market makers ([Nimalendran et al. \(2024\)](#)). [Budish et al. \(2015\)](#) argue that the continuous limit order market design cannot prevent numerous *technical* arbitrage opportunities that are exploitable, fueling a socially wasteful arms race.

The options market possesses unique characteristics that differentiate it from equity markets, likely influencing HFT behavior and its broader market impact. Unlike the order-driven (continuous auction) stock market, the options market operates as a semi-quote-driven system, with quotes predominantly posted by market makers. This structure contributes to wider spreads and slower quote updates, in addition to

exchange-imposed constraints that reflect the illiquidity of options. Moreover, unlike stocks, which can trade across multiple platforms such as dark pools, ECNs, and internalization, listed equity options in the U.S. trade exclusively on exchanges.¹ This exclusivity may influence HFTs’ motivations and strategies in the options market.

In 2009, options exchanges implemented a rule (Regulatory Circular RG09-148) to distinguish high-frequency traders from customers, who do not hold trading permits, such as market makers. Accounts placing an average of 390 or more option orders per day were classified as *professional customers* (PCs). The classification was introduced to promote fair competition and provide regulatory oversight for high-activity accounts, distinguishing them from ordinary customers.² Like traditional market makers (MMs), PCs are permitted to post limit orders on either one or both sides of the market, a privilege not extended to MMs, who cannot post single-sided orders, or to regular customers, who cannot post two-sided orders.³ Furthermore, PCs can leverage their speed to reduce market-making costs. In this study, we use the PC classification as a proxy for identifying high-frequency traders (HFTs) to investigate their role in the options market.

Foucault et al. (2017) provide a theoretical framework to examine toxic arbitrage where HFTs trade against MM’s stale quotes, exacerbating the adverse selection. We extend the model to incorporate both toxic and non-toxic arbitrageurs interacting with MMs with varying speed advantages. Crucially, both arbitrage types widen equilibrium spreads through distinct mechanisms: toxic arbitrage directly imposes adverse selection costs when arbitrageurs exploit stale quotes during price shocks, while non-toxic arbitrage (cream-skimming) forces market makers to widen spreads as compensation for being left with a more toxic residual order flow after arbitrageurs cherry-pick uninformed trades.

Our model yields three key insights. First, option market makers face a trade-off

¹Options wholesalers have abilities to route orders toward a guaranteed allocation proportion if they offer the most competitive price.

²PCs have lower execution priority than customers and are subject to fees for posting trades.

³CBOE Rule 5.51. Market-Maker Obligations, Rule 8.20 Prohibition Against Customers Functioning as Market-Makers.

wherein narrow spreads expose them to being picked off, while wide spreads sacrifice competitive advantage in market making. Second, we demonstrate that speed advantages generate state-contingent effects on equilibrium spreads: in less volatile markets, non-toxic arbitrageurs erode market quality through cream-skimming (convex relationship), while in volatile markets, this effect attenuates as protective spreads dominate (concave relationship). This implies that policies regulating arbitrage speed, such as limiting high-frequency trading, may be more effective in low-volatility environments. Third, we establish that arbitrageur’s expected profits exhibit non-monotonic behavior with respect to volatility due to the equilibrium response of market maker’s optimal spread in equilibrium.

Based on our theoretical model predictions, we examine four empirical channels⁴ that characterize how professional traders operate in options markets: *market-making* (liquidity provision), *cream-skimming* (phantom liquidity), *arbitrage* (liquidity consumption), and *short-run informed trading* (liquidity consumption). The *market-making hypothesis* predicts that HFTs act as alternative liquidity providers, improving market quality by narrowing bid-ask spreads and reducing price impact. Second, the *cream-skimming hypothesis* suggests that HFTs selectively provide liquidity to profitable uninformed order flow, such as routine hedging, while strategically avoiding informed trades. By exploiting speed advantage and fewer regulatory constraints, these traders can cherry-pick and leave adverse selection risk to other market makers. Third, the *arbitrage hypothesis* asserts that HFTs engage in arbitrage thereby increasing adverse selection and widening spreads. Finally, the *informed trading hypothesis* proposes that HFTs leverage private information about stock prices or volatility to execute strategic trades, exacerbating adverse selection costs for market makers and leading to wider spreads and greater price impact.

Since these channels are not mutually exclusive in that PCs may combine arbitrage with selective liquidity provision, we employ a comprehensive empirical approach. To test our

⁴High-frequency trading (HFT) strategies span a range of behaviors. For instance, back-running involves aligning with the direction of informed order flow, while front-running entails establishing positions in advance to trade against anticipated informed orders (e.g., [Yang and Zhu \(2020\)](#); [Xu and Cheng \(2023\)](#)).

hypotheses, we utilize a rich 10-minute-level record of all individual entity-level purchase and sale volumes from the CBOE C1 OpenClose dataset, alongside real-time transactions from the Options Price Reporting Authority (OPRA), covering the period from October 2019 to March 2021. We examine the strategic role of PCs across five key dimensions that collectively distinguish between these potentially overlapping channels: (1) primary trading counterparties, (2) response to the increases in underlying asset volatility, (3) trading activity around earnings announcements, (4) the predictability and profitability of their trades, and (5) their impact on effective bid-ask spreads.

Our initial correlation test strongly indicates that PCs primarily interact with customer order flow and do not exploit market maker quotes. PCs consistently take the opposite side of customer trades; selling when customers buy, buying when customers sell. This is particularly pronounced for large volume trades that contain more than 200 contracts each. The negligible correlation between PC and MM volumes indicates that PCs are not picking off market maker quotes or engaging in arbitrage.

Further, we find that PCs withdraw from providing liquidity when adverse selection risk increases.⁵ While options become more attractive to customers as volatility increases, they simultaneously pose heightened risk for liquidity providers who cannot hedge volatility exposure through the underlying stock. Our logistic and Poisson regressions reveal PCs withdraw liquidity when implied volatility increases, a measure of future volatility. Simultaneously, holding implied volatility constant, they increase participation during periods of elevated historical volatility which leads to wider spreads without an increase in information risk. This pattern is pronounced around earnings announcements when customer volume increases dramatically. While, PCs strategically reduce their trading activity around earnings announcements, leaving market makers to absorb the adverse selection risk.

Next, we test whether professional customers (PCs) trade on short-term information

⁵This is consistent with literature, e.g., [Goldstein and Kavajecz \(2004\)](#) find that liquidity providers withdraw during market downturns and circuit breaker events due to adverse selection risk, widening spreads.

or engage in back-running by examining whether signed option trading volumes predict subsequent short-term returns in the underlying asset. We find no evidence that PC trades have predictive power for either contemporaneous or future stock returns. By contrast, customer trades show strong predictive content: a one standard deviation increase in customer call (put) purchases is associated with concurrent returns of +34% (−18%) and future returns of +38% (−16%) annualized over a 10-minute horizon.

To further isolate the information content by counterparty, we employ a two-stage least squares (2SLS) approach, instrumenting customer volumes separately through PCs and market makers (MMs). The results are striking. A one standard deviation increase in customer call purchases executed against MMs correspond to 69% annualized concurrent returns and a 31% price impact. The same trades executed against PCs, however, show only a 7% concurrent association and a 1.4% price impact. We interpret this selective matching as evidence that PCs strategically avoid toxic order flow, filling only uninformed trades, while market makers are left to absorb the informed (toxic) trades.

Finally, we examine how PC activity affects market quality through its impact on effective spreads. Cream-skimming should leave market makers with more toxic residual order flow, forcing them to widen spreads. The results strongly support this prediction. We test this using two-stage least squares where first-stage regressions instrument PC and MM volumes with customer volumes. PC volume significantly increases effective spreads, with a one standard deviation increase in PC trade volume *raising* spreads by 0.47%. The effect is particularly pronounced for large PC trades, which increase spreads by 0.50%. On the other hand, market maker volume *decreases* spreads by 0.73%.

Our findings suggest that professional customers systematically engage in cream-skimming rather than traditional market making or arbitrage. They mainly trade as customer counterparty, yet avoid customer trades containing information and selectively match with uninformed order flow. Unlike the perceived cream-skimming strategy that targets small retail trades, we observe that PCs are more likely to target large contracts (200 or more),

which likely originate from institutional hedging demands or structured product creation. The evidence demonstrates that PCs extract rents from uninformed flow as market makers absorb adverse selection, substantially widening spreads.

Our study provides novel insights into options market microstructure. Our theoretical model unifies toxic and non-toxic arbitrage mechanisms, establishing that both forms of arbitrage widen spreads and harm market quality, with their relative importance based on volatility. While existing literature primarily focuses on HFTs in equity markets, our research is among the first to examine their behavior in options markets using entity-level data. We also extend the intraday price discovery literature beyond daily and weekly frequencies examined in prior work (e.g., [Chakravarty et al. \(2004\)](#); [Pan and Poteshman \(2006\)](#); [Johnson and So \(2012\)](#); [Bergsma et al. \(2020\)](#)) to 10-minute intervals.

Recent studies document manipulative practices by designated market makers in options markets. [Hendershott et al. \(2022\)](#) demonstrate incentives for options wholesalers to engage in cream skimming through order routing privileges. [Ernst and Spatt \(2022\)](#) document negative impacts of order routing on price improvement. [Bryzgalova et al. \(2023\)](#) show how payment for order flow arrangements enable dominant wholesalers to engage in selective execution practices. Our findings reveal that cream skimming extends beyond designated market makers to all PCs. We find HFTs maximize profits against market design, consistent with [O’Hara \(2015\)](#)’s description of high-frequency trading behavior.

The remainder of this paper is structured as follows. Section [1](#) presents a simple model of trading that generates testable hypotheses and predictions. Section [2](#) details our empirical hypotheses. Section [3](#) provides a description of the data, variables and empirical results. Section [4](#) concludes.

1 Model

This section presents a cross-market arbitrage model following [Foucault et al. \(2017\)](#), in which an underlying asset (denoted stk) and its corresponding option (denoted opt) are traded in segmented markets. A stock market maker (denoted by M_{stk}) sets prices for the stock, while an option market maker (M_{opt}) provides option quotes. An arbitrageur, P , operates only in the options market. Although a no-arbitrage condition links the stock and option prices, the two markets are assumed to be segmented similarly to [Boulatov et al. \(2013\)](#). Due to segmentation, information available to one market maker is not instantly reflected in the quotes of the other market. Competition forces both market makers to earn zero expected profit, while the arbitrageur maximizes gains by exploiting stale quotes (toxic arbitrage) or through cream skimming (non-toxic arbitrage).

1.1 Environment

The model has three dates, $t \in \{0, 1, 2\}$. At date $t = 2$, the payoff to the stock $\tilde{\nu}_s$ is realized. The payoff is given by $\tilde{\nu}_s = \mu_s + \tilde{\epsilon}_s$. At $t = 0$, the stock market maker sets $s_0 = \mu_s$, reflecting the expected stock value at $t = 2$. Simultaneously, the option market maker sets $d_0 = \Phi(s_0, \Psi)$, where Ψ includes characteristics such as strike price, expiration, volatility, and risk-free rate. A price shock to the stock occurs at $t = 1$ with probability α , altering the stock's payoff by $\epsilon_s = \pm \frac{\sigma_s}{2}$ with equal probability. The option's fair value changes by $\pm \frac{\sigma_{opt}}{2}$, where $\sigma_{opt} = \Delta_{opt}\sigma_s$, where Δ_{opt} is the Delta of the option.⁶

Execution Speeds

Speed is a choice variable for traders, and we model it similarly to [Foucault et al. \(2003\)](#) and [Foucault et al. \(2017\)](#). The costs of speed technology reflect resources used for frequent monitoring, rapid order submission, and swift quote adjustments. As traders increase

⁶The expected value and variance of the stock payoff process are derived in Internet Appendix.

their monitoring intensity, they invest more in infrastructure, co-location, and algorithms (see Foucault et al. (2013), Budish et al. (2015), Menkveld (2016)). Consistent with the literature, we assume a cost function that increases with speed. Though a quadratic form often appears in theoretical models to capture increasing marginal costs, we adopt a simpler linear specification for tractability here. Since market makers are obligated to provide liquidity across a broader range of options, arbitrageurs have no such obligation; we posit that the market maker's costs (c_M) are generally higher than those of the arbitrageur (c_P). To update quotes, the M_{opt} invests in a speed technology to change quotes with time τ_M distributed exponentially with parameter λ_M , i.e., $\tau_M \sim \text{Exp}(\lambda_M)$. Similarly, the speed for the Arbitrageur (P) is $\tau_P \sim \text{Exp}(\lambda_P)$. The associated linear costs for the MM and the arbitrageur are: $\mathbb{E}[\tilde{C}_M] = \frac{c_M}{2} \lambda_M$ and $\mathbb{E}[\tilde{C}_P] = \frac{c_P}{2} \lambda_P$. Further, based on the exponential distributions for the speeds, the probability that P is faster than the MM is $\Pr(\tau_P < \tau_M) = \pi = \frac{\lambda_P}{\lambda_P + \lambda_M}$.⁷

If P reacts faster to a shock (probability α), it engages in *toxic arbitrage* by trading on stale option quotes before M_{opt} can adjust her quote. Absent a shock (probability $(1 - \alpha)$), P can *cream skim* the liquidity-driven customer trades by slightly improving upon M_{opt} 's quotes.

Options Market Maker (M_{opt})

At $t = 0$, the options market maker, M_{opt} , provides liquidity by quoting an ask price $a_{opt} = d_0 + \frac{\delta_{opt}}{2}$ and a bid price $b_{opt} = d_0 - \frac{\delta_{opt}}{2}$ conditional on all the information available to her, where d_0 is the option's fair value based on the expected stock price μ_s , and δ_{opt} is the bid-ask spread. The market maker is assumed to be risk-neutral, and competition drives the market maker's expected profits to zero.

Three scenarios influence the M_{opt} 's expected profit: First, if a stock shock occurs. The arbitrageur trades with M_{opt} 's stale quote (with probability $\alpha\pi$), the M_{opt} suffers a loss of

⁷Note that the expected speed (time) for the market maker is $\frac{1}{\lambda_M}$, and $\frac{1}{\lambda_P}$ for the arbitrageur. The proof of the formula can be found in Karlin and Taylor (1975).

$(\frac{\sigma_{opt}}{2} - \frac{\delta_{opt}}{2})$ per trade. Second, when a shock occurs and M_{opt} updates quotes before the arbitrageur trades (probability $\alpha(1 - \pi)$), no toxic arbitrage occurs, and the M_{opt} avoids losses. In this case, the M_{opt} achieves zero net gain or loss from the shock event. Finally, in no-shock states (probability $1 - \alpha$), when trades are due to liquidity demand, the M_{opt} will earn $\frac{\delta_{opt}}{2}$ per trade, unless the arbitrageur undercuts the M_{opt} 's quotes. When undercutting occurs (with probability π), the M_{opt} earns no profit on those trades, leaving only $(1 - \pi)$ of the trades profitable.⁸ Hence, the M_{opt} 's expected profit is:

$$\mathbb{E}[\Pi_M] = -\alpha\pi \left(\frac{\sigma_{opt}}{2} - \frac{\delta_{opt}}{2} \right) + (1 - \alpha)(1 - \pi) \frac{\delta_{opt}}{2} - \frac{c_M}{2} \lambda_M, \quad (1)$$

Arbitrageur (P)

We consider two potential arbitrage strategies: (1) *Toxic arbitrage*, and (2) *Non-toxic arbitrage or cream skimming*. Similarly to the market maker, the arbitrageur invests in speed with rate λ_P , incurring a cost linear in τ_P : $\mathbb{E}[\tilde{C}_P] = \frac{c_P}{2} \mathbb{E}[\tilde{\tau}_P] = \frac{c_P}{2} \lambda_P$.

Toxic Arbitrage Consider the option pricing function $\Phi(s, \Psi)$, where s is the stock price and Ψ includes other parameters such as strike price, time to expiration, risk-free rate, and volatility. After a shock $\tilde{\epsilon}_s \in \{+\frac{\sigma_s}{2}, -\frac{\sigma_s}{2}\}$, the updated option price immediately after the shock, but before the M_{opt} updates at $t = 1^+$ can be approximated as:

$$d_0 + \tilde{\epsilon}_d \approx \Phi(\mu_s + \tilde{\epsilon}_s, \Psi) \approx \Phi(\mu_s, \Psi) + \Delta_{opt} \tilde{\epsilon}_s = d_0 + \Delta_{opt} \tilde{\epsilon}_s,$$

where $\Delta_{opt} = \frac{\partial \Phi}{\partial s}$ evaluated at $s = \mu_s$, i.e., the option's delta at the initial stock price. If the stock experiences a shock of magnitude $\pm \frac{\sigma_s}{2}$, the option price will shift accordingly by $\pm \frac{\sigma_{opt}}{2}$, where $\sigma_{opt} = \Delta_{opt} \sigma_s$. When a shock occurs at $t = 1$ with probability α , the arbitrageur, who moves first with probability $\pi = \frac{\lambda_P}{\lambda_P + \lambda_M}$, can pick off resting quotes before the option

⁸We omit explicit profit for the “shock, but no arbitrage,” viewing it yields zero payoffs for the market maker: no stale-quote losses and no incremental liquidity gains in that brief post-shock window. The detailed derivation of the market maker's profit function is provided in Appendix.

market maker updates them at $t = 2$. Since positive and negative shocks are equally likely, the expected toxic arbitrage profit per trade is $\alpha\pi(\frac{\sigma_{opt}}{2} - \frac{\delta_{opt}}{2})$.

Non-Toxic Arbitrage During non-shock states, $Pr(1 - \alpha)$, the arbitrageur engages in *non-toxic arbitrage* or *cream skimming* by leveraging her speed advantage to trade with liquidity traders by undercutting the market maker's quotes by a small amount $\frac{\eta}{2}$ and earning $(\frac{\delta_{opt}}{2} - \frac{\eta}{2})$.⁹

The probability that the arbitrageur can post and execute these improved quotes before the M_{opt} can respond is the speed-based probability $\pi = \frac{\lambda_P}{\lambda_P + \lambda_M}$. Thus, during non-shock periods, the arbitrageur successfully cream-skims liquidity trades with probability π , earning approximately $\frac{\delta_{opt}}{2}$ per trade. Summing toxic and non-toxic gains and subtracting speed costs gives the expected profit for the arbitrageur, Π_P .

$$\Pi_P \stackrel{\eta \rightarrow 0}{\approx} \pi \left[\alpha \left(\frac{\sigma_{opt}}{2} - \frac{\delta_{opt}}{2} \right) + (1 - \alpha) \frac{\delta_{opt}}{2} \right] - \frac{c_P}{2} \lambda_P \quad (2)$$

Where η is the arbitrageur's quote improvement (assumed infinitesimal, $\eta \rightarrow 0$), c_P is the cost parameter for the arbitrageur's speed, and λ_P is the arbitrageur's monitoring intensity.

The arbitrageur's profit function reflects interactions between speed λ_P , the option spread δ_{opt} , and the shock magnitude σ_{opt} . A higher spread δ_{opt} boosts cream-skimming gains in non-shock periods but reduces toxic arbitrage profits as $\frac{\delta_{opt}}{2}$ gets closer to $\frac{\sigma_{opt}}{2}$. Increasing speed λ_P raises the probability π of arbitrage gains, but it also incurs higher costs $\frac{c_P}{2} \lambda_P$, which can erode net profit. The arbitrageur selects λ_P to balance these trade-offs, while the market maker's spread and speed decisions further influence π .

⁹These trades typically arise from activities unrelated to news or fundamental shifts, such as hedging, portfolio adjustments. Since no new information arrives during these periods, there is no expected subsequent price impact on the stock or the option.

1.2 Equilibrium

In equilibrium, the M_{opt} and the arbitrageur simultaneously choose the option spread δ_{opt} and their speed intensities (λ_M, λ_P) . A competitive equilibrium $(\delta_{opt}^*, \lambda_M^*, \lambda_P^*)$ satisfies the following conditions: (1) the MM's zero-profit condition, $\Pi_M(\delta_{opt}^*, \lambda_M^*, \lambda_P^*) = 0$, and (2) M_{opt} and the arbitrageur choose their respective speed intensities to maximize their profits, namely,

$$\lambda_M^* \in \arg \max_{\lambda_M} \Pi_M(\delta_{opt}^*, \lambda_M, \lambda_P^*), \quad \lambda_P^* \in \arg \max_{\lambda_P} \Pi_P(\delta_{opt}^*, \lambda_M^*, \lambda_P).$$

Market Maker's Zero Profit Condition

The option MM's profit combines losses from toxic arbitrage and gains from no-shock liquidity trades, net of speed costs. Setting this equal to zero gives:

$$\underbrace{(1 - \alpha)(1 - \pi^*) \frac{\delta_{opt}}{2}}_{\text{Liquidity trade profit}} - \underbrace{\alpha \pi^* \left(\frac{\sigma_{opt}}{2} - \frac{\delta_{opt}}{2} \right)}_{\text{Toxic arbitrage loss}} = \underbrace{\frac{c_M}{2} \lambda_M}_{\text{Speed technology cost}} \quad (3)$$

Equation (3) captures how a market maker's decisions on spread δ_{opt} and speed λ_M interact with the arbitrageur's speed λ_P through the probability $\pi^* = \frac{\lambda_P}{\lambda_P + \lambda_M}$. Although a wider spread can increase per-trade revenue from liquidity trades when there is no shock, it also creates higher incentives for the arbitrageur to undercut those quotes (non-toxic arbitrage or cream-skimming). As the spread grows, even a small improvement by the arbitrageur can divert liquidity flow away from the market maker, eroding the maker's expected gains. Moreover, the effect of a wider spread on toxic arbitrage depends on the shock magnitude: if σ_{opt} exceeds the spread significantly, the arbitrageur's pick-off profits remain substantial; conversely, if the spread is large relative to the shock, it might diminish toxic arbitrage incentives but still heighten non-toxic arbitrage threats.

Market Maker's Optimal Speed: λ_M

The first-order condition (FOC) with respect to λ_M from Equation (3) yields:

$$\underbrace{\frac{\lambda_P}{(\lambda_M + \lambda_P)^2}}_{\text{Speed factor}} \times \underbrace{\left[\alpha \left(\frac{\sigma_{opt}}{2} - \frac{\delta_{opt}}{2} \right) + (1 - \alpha) \frac{\delta_{opt}}{2} \right]}_{\text{Payoff in Market Making}} = \underbrace{\frac{c_M}{2}}_{\text{Marginal cost of speed}} \quad (4)$$

Increasing the market maker's speed λ_M reduces the probability $\pi = \frac{\lambda_P}{\lambda_P + \lambda_M}$ that the arbitrageur arrives first, thereby lowering expected toxic and non-toxic arbitrage losses. The optimal speed matches the marginal benefit of raising λ_M to the marginal cost of speed. Hence, the equilibrium spread and speeds reflect a trade-off: a narrow spread reduces the profitability of cream-skimming but might leave the market maker more exposed to toxic arbitrage, whereas a wide spread can deter some toxic picks if the shock is small but amplifies cream-skimming potential and might require higher speed to defend. In equilibrium, the market maker's choices of δ_{opt} and λ_M balance out the arbitrageur's choice of λ_P so that the expected profit is driven to zero.

Arbitrageur's Optimal Speed: λ_P

The FOC for the arbitrageur's optimal speed with respect to λ_P based on Equation (2) yields:

$$\underbrace{\pi(1 - \pi)}_{\text{Speed factor}} \times \underbrace{\left[\alpha \left(\frac{\sigma_{opt}}{2} - \frac{\delta_{opt}}{2} \right) + (1 - \alpha) \frac{\delta_{opt}}{2} \right]}_{\text{Payoff in Arbitrage}} = \underbrace{\frac{c_P}{2} \lambda_P}_{\text{Marginal cost of speed}} \quad (5)$$

The term $\pi(1 - \pi)$ captures the effect of increasing λ_P on the arbitrageur's probability of arriving first. As π moves closer to 1, any additional speed gain produces diminishing improvements in arrival probability. Arbitrage payoff $\alpha \left(\frac{\sigma_{opt}}{2} - \frac{\delta_{opt}}{2} \right) + (1 - \alpha) \frac{\delta_{opt}}{2}$ combines successful pick-off or cream-skimming trades. At the optimal level, the arbitrageur increases λ_P until the marginal benefit of arriving first matches the marginal cost of faster monitoring.

In equilibrium, when both players choose their speeds optimally, the common payoff

terms in the two FOCs (4) and (5) imply

$$\rho^* = \frac{\lambda_P^*}{\lambda_M^*} = \frac{c_M}{c_P}$$

where $\rho = \frac{\lambda_P}{\lambda_M}$ is defined as the relative speed advantage. Once ρ^* is known, the equilibrium probability that the arbitrageur arrives first is

$$\pi^* = \frac{\rho^*}{1 + \rho^*} = \frac{\frac{c_M}{c_P}}{1 + \frac{c_M}{c_P}} = \frac{c_M}{c_M + c_P} \quad (6)$$

The equilibrium profits for the market maker can be computed by substituting the equilibrium spread δ_{opt}^* , the market maker's optimal speed λ_M^* , and the arbitrageur's optimal speed λ_P^* into the profit function, and is given by:

$$\begin{aligned} \Pi_M = & \frac{1}{2} \left[(1 - \alpha)(1 - \pi^*)\delta_{opt}^* - \alpha\pi^*(\sigma_{opt} - \delta_{opt}^*) \right. \\ & \left. - (\pi^*(1 - \pi^*)) \left(\alpha(\sigma_{opt} - \delta_{opt}^*) + (1 - \alpha)\delta_{opt}^* \right) \right] \end{aligned} \quad (7)$$

where π^* represents the equilibrium probability of trade execution.

Optimal Spread(δ_{opt}^*) and Speed(λ_M^*, λ_P^*)

Using π^* together with the M_{opt} 's zero-profit condition from Equation (3) and the first-order condition for speed (4), one can solve for the optimal spread:

$$\delta_{opt}^* = \left(\frac{\alpha \pi^* (2 - \pi^*)}{\pi^* (2 - \pi^*) (2\alpha - 1) + (1 - \alpha)} \right) \sigma_{opt}. \quad (8)$$

The optimal speeds for the market maker and the arbitrageur can be derived from their respective first-order conditions. Given the equilibrium spread δ_{opt}^* , the optimal speed for both agents are:

$$\lambda_M^* = \Gamma \cdot \frac{2c_P}{(c_M + c_P)^2}, \quad \lambda_P^* = \Gamma \cdot \frac{2c_M}{(c_M + c_P)^2} \quad (9)$$

where $\Gamma = \alpha \left(\frac{\sigma_{opt}}{2} - \frac{\delta_{opt}^*}{2} \right) + (1 - \alpha) \frac{\delta_{opt}^*}{2}$.

The market maker's optimal speed depends on the arbitrageur's cost of increasing speed, c_P , because the arbitrageur's cost structure determines the intensity of the speed competition. If c_P is low, the arbitrageur can increase her speed cheaply, forcing the market maker to respond by raising λ_M to avoid being exploited. Conversely, when c_P is high, the arbitrageur faces higher costs for speed, which reduces the market maker's competitive pressure to adjust execution speed. Similarly, the arbitrageur's optimal speed λ_P^* depends on the market maker's cost c_M , as the market maker's cost structure determines how aggressively they will compete in speed.

In equilibrium, the options market maker (M_{opt}) sets δ_{opt}^* to balance gains from non-shock liquidity trades against losses from arbitrage in both shock and non-shock states. The equilibrium spread δ_{opt}^* is unique for values of $\alpha \in (0, 1)$. When $\alpha = 1$, the spread is equal to σ_{opt} due to extreme toxic arbitrage. Additionally, when $\alpha = 0$, the spread is zero due to the absence of competition among MMs, as there is no adverse selection cost. The numerator portion, $\alpha\pi^*(2 - \pi^*)$, captures the frequency with which the arbitrageur successfully wins in both shock and non-shock periods, prompting a wider equilibrium spread to offset those pick-off losses. The denominator reflects how non-shock periods $(1 - \alpha)$ moderate this effect, as the M_{opt} earns half-spread revenues when the arbitrageur fails to undercut her quotes. As $(1 - \alpha)$ grows, the M_{opt} expects more revenue from non-informational trades, which lowers the spread by reducing the need to offset toxic arbitrage losses. Finally, $\sigma_{opt} = \Delta_{opt} \sigma_s$ measures how sensitive the option price is to the underlying shock size: higher σ_{opt} raises the M_{opt} 's potential toxic-arbitrage losses, leading to a larger optimal spread.

Comparative Static Results for Optimal Spread δ^*

The equilibrium spread from Equation (8) exhibits economic trade-offs underlying M_{opt} 's optimal pricing strategy. The signs for the first and second partial derivatives of δ^* w.r.t. α and π^* are summarized below, with derivations provided in the Appendix 7.

With respect to shock probability α , the first derivative $\frac{\partial \delta_{opt}^*}{\partial \alpha}$ is positive, reflecting increasing adverse selection costs as shock frequency rises. The second derivative $\frac{\partial^2 \delta_{opt}^*}{\partial \alpha^2}$ is positive when $2\pi^*(2-\pi^*) < 1$ and negative when $2\pi^*(2-\pi^*) > 1$. With respect to equilibrium probability π^* , the first derivative $\frac{\partial \delta_{opt}^*}{\partial \pi^*}$ is positive, capturing that faster arbitrageurs impose greater losses through toxic arbitrage and cream-skimming. The second derivative $\frac{\partial^2 \delta_{opt}^*}{\partial (\pi^*)^2}$ is positive for $\alpha < 0.5$ and negative for $\alpha > 0.5$.

[Figure 1]

Figure 1 illustrates how the relationship between δ_{opt}^* and π^* transitions from convex to concave as α increases. For low α , M_{opt} maintains narrow spreads due to limited adverse selection risk, but spreads widen at an accelerating rate as π^* rises and the arbitrageur becomes more effective at exploiting stale quotes. For high α , frequent price shocks widen baseline spreads, and the marginal impact of increasing π^* diminishes since adverse selection costs already dominate pricing decisions.

When M_{opt} 's speed cost c_M is low, investing in speed becomes economical, reducing the arbitrageur's advantage and enabling narrower spreads. Conversely, high c_M discourages speed investment but requires wider spreads to mitigate risks. These results reveal that for low-volatility stocks, spreads are primarily influenced by π^* , while for high-volatility stocks, spreads are dominated by α . This implies that policies regulating arbitrage speed, such as limiting high-frequency trading, may be more effective in low-volatility environments.

Arbitrageur's Expected Profit

Equilibrium expected profits for the arbitrageur are derived by substituting the equilibrium spread δ_{opt}^* and the equilibrium π^* in the arbitrageur's profit function:

$$\begin{aligned} \mathbb{E}[\Pi_P] = & \frac{1}{2} \left[\alpha \pi^* (\sigma_{opt} - \delta_{opt}^*) + (1 - \alpha) \pi^* \delta_{opt}^* \right. \\ & \left. - (1 - \pi^*) \pi^* (\alpha (\sigma_{opt} - \delta_{opt}^*) + (1 - \alpha) \delta_{opt}^*) \right] \end{aligned} \quad (10)$$

[Figure 2]

Figure 2 describes the relationship between the arbitrageur’s profit and α modulated by π^* . Specifically, at low levels of volatility, profit increases with α because toxic arbitrage opportunities expand. However, the M_{opt} responds defensively by widening the spread for higher values of α (Equation 8 and Figure 1), thereby diminishing arbitrage gains. A flat pattern when the speed advantage is relatively low indicates that arbitrageurs cannot effectively capitalize on increased volatility due to market makers’ reactive spread adjustments. This suggests that option arbitrageurs’ participation in the market is not necessarily characterized by a significant negative association with increased volatility, particularly when speed advantage is low. Thus, the model predicts that their trading participation does not always increase with volatility, as market makers’ defensive adjustments can counteract the potential gains from heightened price fluctuations.

1.3 Model Summary

Our model captures the strategic interaction between an option market maker (M_{opt}) and an arbitrageur (P) in segmented markets where speed is determined endogenously. The central tension arises from two competing arbitrage mechanisms: *toxic arbitrage* during price shocks (probability α), and *cream-skimming* during non-shock periods (probability $1 - \alpha$). When shocks occur, the arbitrageur exploits stale quotes before the market maker can adjust. In contrast, during quiet periods, cream-skimming enables the arbitrageur to undercut the market maker’s quotes and capture liquidity-driven trades. The risk-neutral option market maker sets the equilibrium spread to balance expected losses from arbitrage against profits from liquidity trades.

The model predicts that greater arbitrageur speed advantage (π^*) and higher shock frequency (α) both lead to wider spreads, as adverse selection costs dominate. However, the relationship exhibits nonlinearities depending on which arbitrage strategy dominates.

In low-volatility environments ($\alpha < 0.5$), arbitrageurs primarily engage in cream-skimming by selectively trading with uninformed customers, which forces market makers to face a higher proportion of informed traders. This creates a convex relationship where speed improvements have increasing effects on widening the spread. Conversely, in high-volatility settings ($\alpha > 0.5$), toxic arbitrage dominates as frequent shocks create opportunities to exploit stale quotes. The relationship becomes concave because spreads are already wide to accommodate baseline adverse selection risks, thereby diminishing the marginal effect of additional speed advantages.

2 Empirical Implications of the Model

The theoretical framework developed above generates several testable predictions about the relationship between professional customer (PC) trading activity and option market dynamics. Our model’s prediction that arbitrageurs (PCs) strategically choose between toxic arbitrage and cream-skimming offers testable hypotheses to interpret diverse empirical patterns in high-frequency options trading.

The equilibrium condition $\pi^* = \frac{c_M}{c_M + c_P}$ reveals that the arbitrageur’s speed advantage depends solely on relative cost structures rather than absolute trading volumes. However, empirically proxying this latent speed parameter directly is challenging. Due to the strategic nature of order placement, PCs are not obligated to trade or provide quotes at all times. Specifically, observed trading volumes capture both (1) the intent to trade and (2) the *realized* speed advantage relative to MMs. We use PC trading volume share as a proxy for realized speed advantage, which represents the lower bound of successful preemptive trading before market makers can update quotes.

Our model’s dual arbitrage mechanisms provide a framework for evaluating several hypotheses, and Table 1 summarizes the testable predictions for these hypotheses.

[Table 1]

2.1 H1: Liquidity Provision Hypothesis

The liquidity provision hypothesis posits that HFTs act as beneficial market participants, enhancing liquidity without harming traditional market makers. [Hendershott et al. \(2011\)](#); [Hendershott and Riordan \(2013\)](#) characterize HFTs as liquidity providers that both absorb and supply liquidity. The literature predominantly documents positive relationships between high-frequency trading and market quality, evidenced by tighter bid-ask spreads and enhanced price efficiency [Carrion \(2013\)](#); [Jones \(2013\)](#); [Menkveld \(2013\)](#); [Brogaard et al. \(2014\)](#); [O’Hara \(2015\)](#), suggesting that HFTs complement rather than compete destructively with designated market makers.

Under H1, PCs supply liquidity as counterparties to customer demand, particularly during high (implied) volatility and earnings announcements, similar to market makers. This liquidity provision should narrow bid-ask spreads, reduce price impact, and enhance market efficiency. We expect negative correlations between PC volume and both option spreads and stock price impact.

2.2 H2: Cream Skimming Hypothesis

The cream-skimming channel is consistent with the “phantom liquidity” hypothesis, where PCs selectively provide liquidity to uninformed trades, while avoiding adverse selection during volatile periods. [Easley et al. \(1996\)](#) document that payment-for-order-flow agreements are consistent with this behavior, as proprietary trading firms use algorithms to supply liquidity mainly to uninformed orders while avoiding high information asymmetry periods([Bergsma et al. \(2020\)](#)). In options markets, where significant trading occurs within the NBBO, HFTs can strategically execute against resting orders and skim uninformed orders using proprietary algorithms ([O’Hara \(2015\)](#)), intensifying adverse selection risk for MMs ([Kyle \(1985\)](#); [Glosten and Milgrom \(1985\)](#)). When α is low (few shocks), our model predicts that PC activity should concentrate on capturing the spread through non-toxic

arbitrage, consistent with cream-skimming behavior that increases market maker adverse selection costs. This selective participation by PCs forces designated market makers to handle more toxic order flow. As a result, spreads widen and price impact increases (Easley et al. (2012)). Options market makers face additional regulatory burdens including penalties for frequent quote adjustments (Muravyev and Pearson (2020)) and higher monitoring costs (Foucault et al. (2003)). These constraints disadvantage them relative to agile algorithmic traders.

Under H2, since PCs supply liquidity selectively, they will predominantly trade with customers rather than market makers. If PCs target uninformed traders, they will trade less before earnings announcements or anticipated uncertainty compared to normal market conditions. Increased PC market participation should lead to higher effective spreads, reflecting the adverse selection risk MMs face when left with more toxic order flow.

2.3 H3: Arbitrage Hypothesis

The toxic arbitrage channel corresponds to arbitrage strategies where PCs exploit stale quotes following information shocks. This hypothesis is motivated by regulatory changes where options exchanges eliminated HFT priority, forcing them to engage in more opportunistic strategies.¹⁰ Foucault et al. (2017) demonstrated that arbitrage strategies utilizing speed advantages to pick off stale quotes can be toxic to market makers. The channel is supported by frequent put-call parity violations across exchanges Andersen et al. (2021), with prior research by Budish et al. (2015) and Shkilko and Sokolov (2020) showing that HFTs can profitably exploit such opportunities. Our comparative statics in Figure 2b show that beyond a certain threshold, higher shock frequency (α) amplifies the profitability of toxic arbitrage. Intuitively, more frequent shocks provide additional opportunities for traders to exploit stale quotes, thereby amplifying gains from this strategy.

¹⁰After the Regulatory Circular RG09-14, there was concern that the rule would force HFTs to be more opportunistic, saying, "You are forcing the customer to adopt a more active trading strategy to pick off the market maker."

Under H3, PCs frequently trade when short-run arbitrage opportunities arise, particularly during volatile periods when high (implied) volatility in underlying stocks can cause MM quotes to become stale. PCs demand liquidity and immediacy and trade against stale quotes by MMs, leading to high correlation between PC and MM trading volumes. If PC trading volume rises, market makers are likely to widen bid-ask spreads to compensate for the risk of trading at stale quotes.

2.4 H4: Short-run Informed Trading Hypothesis

The informed trading hypothesis posits that PCs possess private information regarding underlying stocks' future direction or volatility. The embedded leverage in options incentivizes informed strategic trading [Easley et al. \(1998\)](#); [Kyle \(1985\)](#). Given options' lower liquidity and higher transaction costs, informed traders who employ high-frequency strategies mitigate short-lived information leakage through small, rapid executions (e.g., [Hasbrouck and Saar \(2009\)](#)). Also, options' ability to trade on volatility leads to trading patterns aligned with option Greeks, such as increased activity in high-vega options during periods of heightened uncertainty [Bates \(2000\)](#); [Ni et al. \(2008\)](#).

Under H4, if PCs possess directional information, their positions should predict short-run price movements; long call (long put) predicting positive (negative) underlying stock price movements, and volatility positions as straddle positions predicting future volatility. Also, informed PCs would trade likely with market makers (demanding liquidity) to ensure immediate execution. PC activity should widen bid-ask spreads as market makers compensate for adverse selection. Trading volume should increase before earnings announcements and other information events.

3 Empirical Results

3.1 Data and methodology

Our dataset is constructed from the Chicago Board Options Exchange (CBOE), the Center for Research in Security Prices (CRSP), the New York Stock Exchange Trades and Quotes (TAQ), the Thomson Reuters Institutional Brokers Estimate System (I/B/E/S), and the U.S. Department of Treasury. We focus on equity options for S&P 500 firms that were continuously listed during our sample period (October 2019 to March 2021) using CRSP data. We incorporate Treasury yield curve data interpolated at the daily level to match exact days to expiration, dividend yields from CRSP, historic volatility measures computed from high-frequency TAQ data at 10-minute intervals, and VIX quotes matched using the most recent available information.

Our primary CBOE data comprises two products: C1 data providing detailed transaction volumes by trader type at 10-minute intervals, and OPRA data containing all trades and quotes across 16 U.S. options exchanges (January 2019 to December 2021). C1 data include open/close position indicators for each trader category and are aggregated to provide cumulative snapshots of signed option trading volumes by trader type: Customers, PCs¹¹, and MMs.¹² The C1 data further categorizes customer and professional customer trades by contract size: small (fewer than 100 contracts), medium (100-199 contracts), and large (200 or more contracts). We first-difference the cumulative volumes to obtain net trading volume per 10-minute interval, up to 4:00 p.m., EST. To estimate MMs' open interest (OI) at 10-minute intervals, we identify new contracts by requiring their earliest trading record to be after the beginning of our sample period. Following [Ni et al. \(2021\)](#), we measure

¹¹A PC is a person or entity that engages in highly frequent options trading that is not a broker or dealer in securities, and places more than 390 orders in listed options per day on average during a calendar month for their beneficial account(s).

¹²We limit our focus to those entities and exclude others such as brokers and clearing firms.

MMs' net exposure using $NetOpenInterest_{j,t}$:

$$NetOpenInterest_{j,t} = - \sum_{Y \in \text{Non MM}} \{ (OpenBuy_{j,t}^Y - CloseSell_{j,t}^Y) - (OpenSell_{j,t}^Y - CloseBuy_{j,t}^Y) \}$$

where Y represents non-MM trader types. $OpenBuy$ and $OpenSell$ establish new positions, while $CloseBuy$ and $CloseSell$ offset existing positions. This represents marginal option volume MMs must hedge at time t . We proxy marginal delta hedging cost by weighting this with the option's gamma $\Gamma_{j,t}$ with the underlying stock's bid-ask spread.

Transaction-level information from OPRA, including bid-ask spreads on options and underlying stocks, Greeks, and implied volatility, is aggregated at 10-minute intervals using volume-weighted averages to match C1. We calculate implied volatility and Greeks (delta, gamma, vega, theta) using Black-Scholes for each transaction. Implied volatility is derived from midpoint of the NBBO prices for stocks and options, strike prices, time to expiration, CRSP dividend yields, and interpolated Treasury rates. Underlying assets' 10-minute realized volatility is estimated from 1-second return standard deviations, with 10-minute returns calculated using end-of-interval mid-quotes. Market return and volatility are measured from SPY ETF from TAQ. Approximately 30% of OPRA data match our C1 data, consistent with CBOE's share of U.S. options trading. We exclude non-standard options (FLEX options) and index options, intervals with no entity trading, and observations with data quality issues: zero or negative bid-ask spreads, negative volume, negative gamma, or positive theta. Our final sample consists of approximately 13 million observations.

3.2 Preliminary Analysis

[Table 2]

Table 2 presents summary statistics for the variables employed in our analysis, based on 10-minute frequency data encompassing all single-leg options on stocks in the S&P 500 traded on the CBOE. Panel A reports the summary statistics for option volumes by trader

identity, option spreads, implied volatility, and option Greeks. We report aggregated volumes by entity only for brevity.

As MM’s hedge rebalancing affects stock returns (Ni et al. (2021)) we formulate two control variables delta-hedging by MMs: GammaExp and RebalCost. GammaExp is a proxy variable for MMs’ exposure to delta neutrality, constructed by multiplying NetOpenInterest and option Gamma (Γ). This is similar to Net Gamma in Ni et al. (2021) but individual series level. The rebalancing cost variable is the product of Gamma, stock spread, and absolute NetOpenInterest. This measure reflects the cost incurred by market makers when adjusting their delta-hedged positions, as higher Gamma and wider stock spreads increase the cost of maintaining delta neutrality. The volume variables indicate each trader’s put/call aggregated option contracts for each 10-minute interval. Customers transact on average 13.99 contracts during a 10-minute interval, closely matching the market makers’ average volume of 13.93 contracts. Professional customers trade only 0.43 contracts, and the distribution is highly skewed, reflecting their highly selective participation in the options market. Panel B reports underlying stock variables that are matched to our CBOE data: stock and market returns over 10-minute and daily intervals, realized volatility, and relative bid-ask spreads.

[Figure 3]

Figure 3 displays the time trend of customer and professional trader transaction volume by moneyness during the sample period.¹³ Customer trading is dominated by ATM options throughout the period, with notable volume increases during the COVID-19 market disruption in early 2020. Professional traders exhibit more volatile trading patterns with dramatic spikes in ATM and OTM options during 2021. This reflects their highly selective participation. ITM options maintain consistently low volumes for both trader types.

[Figure 4]

¹³For Figure 3, we report based on both existing and new option contracts transactions. For a broader perspective on options and stock trading volume trends across all U.S. exchanges, see Figure A1 in Appendix.

Figure 4 shows total trading volume by moneyness and trade direction. Customers exhibit a strong preference for call options over puts, with ATM options dominating across all trade types at approximately 60-64%. PCs show distinct strategic patterns, concentrating on OTM selling activity: 54.4% for calls and 41.3% for puts. These proportions substantially exceed customers' OTM selling activity of 31% for calls and 36.2% for puts.

Effective and quoted spreads by moneyness

[Table 3]

Table 3 reports summary statistics for options spreads by moneyness groups. At-the-money (ATM) options account for 7.7M observations (59%), out-of-the-money (OTM) options for 4.6M observations (35%), and in-the-money (ITM) options for 0.9M observations (7%). OTM options exhibit the highest quoted relative spread at 17.8% and proportional effective spread at 7.83%. ATM options have a quoted relative spread of 8.82% and an effective spread of 4.26%, while ITM options show the lowest spreads with a relative quoted spread of 5.01% and an effective spread of 1.82%. The dollar quoted spread OTM options is \$0.18, comparable to ATM options of \$0.20, despite their substantially lower option prices. The dispersion of spreads is highest for ITM options.

[Figure 5]

Figure 5 displays trading volume by expiration across moneyness categories. PCs and customers exhibit inverse preferences for ATM options: customers use very short-term contracts (<7 days), while PCs peak in medium-term positions (<30 days). This pattern reverses for longer expirations, suggesting PCs avoid high-gamma, high-theta environments favored by customers. Conversely, both trader types show proportionally similar distributions in ITM options across all expiration periods. A significant gap emerges in OTM short-term options (<7 days), which indicate retail preference for lottery-like

payoffs in short maturity, out-of-the-money contracts, while professionals largely avoid this segment.

Correlations

Panel A in Table 4 shows the correlation coefficients among signed option volumes by trader identity. Not surprisingly, Customer-MM correlations are the highest observed, with customer buy-MM sell at 41% and customer sell-MM buy at 42%. PC correlations are notably comparable to these levels, with customer buy-professional sell at 36% and customer sell-professional buy at 38%, indicating that professionals function as significant liquidity providers to customers alongside market makers. Professional-MM correlations remain minimal (0-3%), suggesting professional trades are unlikely to be hitting MM quotes (demand liquidity) and instead represent independent liquidity provision rather than engaging in arbitrage activities.

[Table 4]

Panels B and C further reveal asymmetric liquidity provision patterns. Professional call selling correlates more strongly with customer call buying (39%) than put selling with customer put buying (29%). Conversely, professional put buying shows higher correlation with customer put selling (37%) compared to call buying with customer call selling (38%).

Trading Around Earnings Announcements

[Figure 6]

Figure 6 presents trading patterns around earnings announcement days (EADs).¹⁴ Panel (a) shows customer volume peaks at EADs (14.8 contracts), whereas professional volume exhibits the opposite pattern, declining from pre-announcement levels to minimal activity

¹⁴If the earnings announcement was before 4PM on trading day t , the EAD is day t . If the earnings is announced after market close on day t , then the EAD is the next trading day.

on EAD. Panel (b) shows that historic volatility peaks on EADs at nearly 1.0, compared with a baseline of 0.66-0.68 in non-announcement periods. Panel (c) displays implied volatility which increase steadily from EAD-3 (0.57) to a peak on EAD-1 (0.64), then declines sharply after the announcement to 0.43 by EAD+3. This is consistent with MMs pricing options for potential jumps in the stock price following the earnings announcements.

3.3 Regressions

Information Content of Customer and Professional Trading

We use underlying stock market excess returns to test the directional information content of customer option trading. We employ a regression model to determine whether customer option volume predicts concurrent and future stock returns using the following fixed-effects regression specification:

$$r_{i,t+m} = \beta_1 V_{i,j,t}^{BuyCall,C} + \beta_2 V_{i,j,t}^{SellCall,C} + \beta_3 V_{i,j,t}^{BuyPut,C} + \beta_4 V_{i,j,t}^{SellPut,C} + X' \beta + \Lambda_t + \Omega_i + \epsilon_{i,j,t} \quad (11)$$

where i is the underlying stock, j is the option series, and t is the 10-minute time identifier. $r_{i,t+m}$ denotes the underlying excess market return ($r_{i,t} - r_{mkt,t}$) in percentage terms and annualized from 10-minute returns. For concurrent returns ($m = 0$), the return corresponds to the same 10-minute time bucket as the options trading. For future returns, we measure subsequent stock performance using both forward returns in later intervals and cumulative holding-period returns.

$V_{i,j,t}^{BuyCall,C}$ is customer's standardized buy transaction volume for call option j at time t for underlying equity i . The other option volume variables are similarly defined. The vector X includes control variables: lagged excess returns (10- and 20-minute lags, to control for momentum effects), realized volatility, market volatility, rebalancing cost, and gamma exposure. We control for the delta hedging channel through rebalancing cost and gamma exposure variables to isolate directional information from mechanical hedging effects. The

Λ_t and Ω_i are time fixed effects and option-day fixed effects, respectively. Standard errors are two-way clustered by option-day and time.

[Table 5]

Our estimation results are presented in Table 5. Customer buy positions in both calls and puts are strongly aligned with their directional expectations and contain meaningful information about underlying stock movements. A one-standard-deviation increase in customer buy-call volume associates with 33.90% annualized 10 minute excess returns and 38.09-40.58% higher annualized holding returns over 10-20 minutes. However, customer trading does not explain future stock returns beyond contemporaneous movements. The information content arises primarily through contemporaneous price discovery, suggesting that the predictive power may operate even shorter than our 10-minute measurement interval. Only customer sell put volume demonstrates limited significance for 20-minute forward returns (3.585%), while other specifications show no meaningful future predictability.

Leveraging our comprehensive entity-level trade volume data, we employ a two-stage least squares approach to identify which counterparties absorb informed customer flows. We decompose customer trading based on counterparty identity using the following specification:

$$V_{i,j,t}^{D,C} = \sum_m \gamma_m V_{i,j,t}^{m,Y} + X' \beta + \Lambda_t + \Omega_i + \zeta_{i,j,t} \quad (12)$$

$$r_{i,t+m} = \beta_1 \hat{V}_{i,j,t}^{BuyCall,C} + \beta_2 \hat{V}_{i,j,t}^{SellCall,C} + \beta_3 \hat{V}_{i,j,t}^{BuyPut,C} + \beta_4 \hat{V}_{i,j,t}^{SellPut,C} + X' \beta + \Lambda_t + \Omega_i + \epsilon_{i,j,t} \quad (13)$$

where D represents trade direction and option type combinations, and Y identifies counterparty trader types (professional customers or market makers). Equation 12 specifies the first-stage regression, while equation 13 employs fitted values in the second stage.

[Table 6]

The 2SLS results in Table 6 reveal distinct patterns based on counterparty identity. When customer volumes are fitted with professional customer counterparties (columns 1-2),

we observe minimal information content. Customer buy call volume shows weak significance (7.004% for concurrent returns), while other coefficients remain largely insignificant. These results suggest that professional customers avoid trading against informed customer flows.

Conversely, when customer volumes are fitted with market maker counterparties (columns 3-4), the information content increases substantially. Customer buy call volume demonstrates strong significance with 69.23% association with concurrent returns—more than double the OLS estimate. Similarly, customer buy put volume shows strong negative association (-27.50%) with concurrent returns.

These findings indicate that market makers, bound by liquidity provision obligations, absorb informed customer flows while professional customers strategically avoid such trades. The amplified coefficients in the market maker specifications suggest that when market makers serve as counterparties, the information content of customer trading becomes more pronounced, consistent with adverse selection costs in option market making. We found the information content concentrates primarily in at-the-money and out-of-the-money options ([A1](#), [A2](#), [A3](#) in Appendix).

[Table [7](#)]

We apply similar regression specifications to examine professional-customer trading patterns. Across most specifications, professional-customer trading shows no significant directional information content. Professional volumes exhibit minimal statistical significance in both OLS and 2SLS models, in sharp contrast to customer patterns. This difference indicates that professional customers primarily engage in liquidity provision, arbitrage, or hedging activities rather than information-driven directional trading.

Price Impact

We examine the price impact of customer option trading as a complement to the previous analysis. Following [Amihud \(2002\)](#), we use contemporaneous absolute stock returns as

the dependent variable to measure price impact. Option returns are subject to various drivers—including volatility and time decay—that may obscure underlying information effects, whereas absolute stock returns provide standardized measures that are comparable across different contracts.

[Table 8]

Table 8 presents price impact estimates across different counterparty specifications. The baseline OLS results (column 1) show significant price impact for all customer trading activities. Customer buy call and buy put volumes exhibit the largest effects, with 13.19% and 13.14% absolute annualized return impacts, respectively, for a one standard deviation increases in volume.

The 2SLS decomposition reveals stark differences in price impact based on counterparty identity. When customer volumes are fitted with professional customer counterparties (column 2), one standard deviation increase of customer volume has largely insignificant price impact. Only customer sell put volume shows weak negative significance (-3.953%), while other coefficients lose statistical significance entirely.

Conversely, when customer volumes are fitted with market maker counterparties (column 3), price impacts amplify substantially. Customer buy call volume generates 30.60% absolute return impact, more than double the baseline estimate. Customer sell call and buy put volumes show similarly elevated impacts (23.08% and 21.53% respectively). Even customer sell put volume, the weakest signal in baseline specifications, demonstrates significant impact (13.29%) when transacted with market makers. These patterns indicate that market makers bear substantial adverse selection costs against informed customer order flow, whereas PCs appear to successfully avoid trades that generate meaningful price impact.

Professional Customer Trading and Volatility

We test whether PCs absorb heightened volatility risk, like MMs, or instead strategically withdraw. Unlike designated MMs bound by exchange rules and obligations, PCs are not

obligated to make markets. Their participation is strategic and highly selective, and they may choose not to participate under certain market conditions. This strategic behavior is evidenced by the substantial number of zeros in trading activity compared with customers and MMs in Table 2.

We address this structure using a weighted Poisson to properly model both participation choice and trading intensity given their participation. We first estimate logistic regression for $P(I(Y > 0)|X)$ to analyze participation probability. We then use these participation probabilities as observation weights in Poisson regression to isolate the intensity effect.¹⁵

[Table 9]

Table 9 show PCs systematically withdraw before periods of heightened future uncertainty. A one standard deviation increase in implied volatility reduces participation odds by 18.4% overall ($e^{-0.203} = 0.816$) and 19.1% for ATM options ($e^{-0.212} = 0.809$). This withdrawal pattern intensifies around earnings announcements: PCs reduce participation by 46.2% the day before ($e^{-0.620} = 0.538$) and 47.0% on announcement day ($e^{-0.634} = 0.531$). Together, high implied volatility and earnings events signal uncompensated risk that PCs avoid. Instead, high implied volatility attracts PCs to trade ITM options, increasing their participation odds by 15.7% ($e^{0.146} = 1.157$), as these contracts draw less informed order flow and require minimal hedge rebalancing.

Historic volatility shows opposite effects. After controlling for implied volatility¹⁶, historic volatility represents predictable volatility, which can create profitable market-making through widened spreads.

Our results also show strategic flow selection through order size. Medium orders (>100 contracts) increase PC participation by 97% ($e^{0.678} = 1.97$). While large orders (>200

¹⁵This approach is similar to a zero-inflated Poisson model. We use weights to consistently report large panel regression coefficients with high-dimensional fixed effects and robust clustered standard errors. Our log-regression results yield unchanged interpretations.

¹⁶We observe substantial cross-sectional variation between underlying stock volatility and option-specific implied volatility, since our data is individual option series level.

contracts) raise the odds by 293.9% ($e^{1.371} = 3.939$). These findings suggest that PCs target non-informational institutional flows.¹⁷ We report detailed results by signed volume and moneyness in Appendix.

[Table 10]

Historic volatility generates similar asymmetry. Customers uniformly increase trading by 7.8% overall ($e^{0.0749} = 1.078$), reaching 15.9% for ATM ($e^{0.1472} = 1.159$). PCs exhibit minimal response at 3.5% ($e^{0.0340} = 1.035$) overall, with insignificant effects across moneyness categories. Earnings announcements reveal the starkest liquidity mismatch. Customers increase trading on announcement day by 7.8% ($e^{0.0748} = 1.078$) overall and 10.3% ($e^{0.0977} = 1.103$) for ATM options. In contrast, PCs show no significant main effects on announcement days, consistent with their withdrawal documented in the logistic regression.¹⁸ Further Poisson regression results for customers and PC are reported in the Appendix.

Impact on Effective Spreads

Our theory predicts that cream skimming by professional customers widens equilibrium spreads through adverse selection. We test this prediction using effective spreads based on actual transactions and quotes.

We present two stage results to isolate each counterparty’s impact on effective spreads and OLS as baseline. We estimate the following:

$$V_{i,j,t}^{MM}, V_{i,j,t}^P = \sum_m \gamma_m V_{i,j,t}^{m,C} + X'\beta + \Lambda_t + \Omega_i + \zeta_{i,j,t} \quad (14)$$

$$Spread_{i,j,t} = \beta_1 \hat{V}_{i,j,t}^{MM} + \beta_2 \hat{V}_{i,j,t}^P + X'\beta + \Lambda_t + \Omega_i + \epsilon_{i,j,t} \quad (15)$$

¹⁷Institutions use options for risk management, and mandate compliance, employing strategies like hedging with puts/collars, harvesting carry via covered calls or cash-secured puts. These flows are mechanical, and liquidity seeking but not signal-based.

¹⁸We observe higher ATM trading intensity for PCs in the EAD-1 period, suggesting a mitigated impact before the EAD.

Since PCs selectively participate while MMs mostly make the market, we examine how PC trading participation affects effective spreads by their trade sizes as well. We use dummy variables to capture both their participation and size effects on spreads:

$$Spread_{i,j,t} = \beta_1 \hat{V}_{i,j,t}^{MM} + \sum_s \beta_{2,s} (\hat{V}_{i,j,t}^{MM} \times D_{i,j,t}^s) + X' \beta + \Lambda_t + \Omega_i + \epsilon_{i,j,t} \quad (16)$$

where i indexes underlying assets, j indexes option series, and t represents time. $D_{i,j,t}^s$ are dummy variables for PC size categories. Small indicates any professional volume greater than 0 contracts, Medium indicates volume greater than 100 contracts, and Large indicates volume greater than 200 contracts. Under the cream skimming hypothesis, larger PC trades should have stronger effects on spreads because market makers face more adverse selection when PCs target larger uninformed order flows.

[Table 11]

Table 11 presents results.¹⁹ The cream skimming effects vary substantially across moneyness categories. For ATM options, the two-stage estimates show the strongest effects. Professional volume increases effective spreads by 0.51 percentage points, while market maker volume decreases spreads by 0.84 percentage points. Large professional trades drive most of the spread widening effect, increasing spreads by 0.55 percentage points in the two-stage specification. OTM options show intermediate effects. Professional volume increases spreads by 0.21 percentage points, with large professional trades contributing 0.34 percentage points. The market maker effect is weaker but still negative at 0.23 percentage points.

ITM options exhibit distinctly different patterns. We observe weak significance on PC volume effects, and more notably, the interaction effects reverse sign. Medium and large professional trades actually reduce spreads by 0.27 and 0.04 percentage points.

¹⁹Regression results for effective spreads across the full sample are reported in the Appendix.

The moneyness differences align with adverse selection. ATM options face the greatest adverse selection because they are most sensitive to price movements and are likely to attract informed trading. ITM options have the least concerns regarding adverse selection and PCs can actually enhance market liquidity.

4 Conclusion

Our theoretical model extends [Foucault et al. \(2017\)](#) to examine both toxic and non toxic arbitrage in options markets. The model predicts that arbitrageurs' speed advantages create state-contingent effects on spreads—convex in low-volatility regimes where non-toxic arbitrage dominates, and concave in high-volatility regimes where toxic arbitrage prevails. Our empirical results show that PC participation is highest in low-volatility environments, consistent with the non-toxic arbitrage mechanism.

Our empirical analysis establishes four key facts about professional customer trading. First, PCs trade primarily with customers rather than market makers, acting as selective liquidity providers. Second, PCs systematically avoid directionally informed flow: the price impact of customer trades routed to PCs is significantly smaller than when routed to MMs. Third, PCs reduce participation ahead of periods of anticipated volatility, avoiding trading before known information events. Fourth, PC trading increases effective spreads, consistent with the cream-skimming hypothesis.

The evidence indicates that professional customers operate as sophisticated intermediaries who extract rents from the most profitable uninformed opportunities while leaving market makers to absorb adverse-selection risk. Although PCs provide liquidity, their selective participation ultimately reduces market quality by segmenting order flow and inducing wider spreads for the remaining participants.

Our findings contribute to the literature on high-frequency trading in options markets, offering new evidence on how speed advantages and cream-skimming shape liquidity, adverse

selection, and price discovery.

References

- Amihud, Y. (2002). Illiquidity and stock returns: cross-section and time-series effects. *Journal of Financial Markets* 5(1), 31–56.
- Andersen, T., I. Archakov, L. Grund, N. Hautsch, Y. Li, S. Nasekin, I. Nolte, M. C. Pham, S. Taylor, and V. Todorov (2021). A descriptive study of high-frequency trade and quote option data. *Journal of Financial Econometrics* 19(1), 128–177.
- Bates, D. S. (2000). Post-’87 crash fears in the sp 500 futures option market. *Journal of Econometrics* 94(1-2), 181–238.
- Bergsma, K., A. Fodor, V. Singal, and J. Tayal (2020). Option trading after the opening bell and intraday stock return predictability. *Financial Management* 49(3), 769–804.
- Boulatov, A., T. Hendershott, and D. Livdan (2013). Informed trading and portfolio returns. *The Review of Economic Studies* 80(1), 35–72.
- Brogaard, J., T. Hendershott, and R. Riordan (2014). High-frequency trading and price discovery. *The Review of Financial Studies* 27(8), 2267–2306.
- Brunnermeier, M. K. and L. H. Pedersen (2005). Predatory trading. *The Journal of Finance* 60(4), 1825–1863.
- Bryzgalova, S., A. Pavlova, and T. Sikorskaya (2023). Retail trading in options and the rise of the big three wholesalers. *Journal of Finance* 78(6), 3465–3514.
- Budish, E., P. Cramton, and J. Shim (2015). The high-frequency trading arms race: Frequent batch auctions as a market design response. *The Quarterly Journal of Economics* 130(4), 1547–1621.
- Carrion, A. (2013). Very fast money: High-frequency trading on the nasdaq. *Journal of Financial Markets* 16(4), 680–711.
- Chakravarty, S., H. Gulen, and S. Mayhew (2004). Informed trading in stock and option markets. *The Journal of Finance* 59(3), 1235–1257.
- Easley, D., N. M. Kiefer, and M. O’Hara (1996). Cream-skimming or profit-sharing? the curious role of purchased order flow. *The Journal of Finance* 51(3), 811–833.
- Easley, D., M. M. López de Prado, and M. O’Hara (2012). Flow toxicity and liquidity in a high-frequency world. *The Review of Financial Studies* 25(5), 1457–1493.
- Easley, D., M. O’Hara, and P. S. Srinivas (1998). Option volume and stock prices: Evidence on where informed traders trade. *The Journal of Finance* 53(2), 431–465.
- Ernst, T. and C. S. Spatt (2022). Payment for order flow and asset choice. NBER Working Paper. No.

- 29883, Available at: <https://www.nber.org/papers/w29883>.
- Foucault, T., O. Kadan, and E. Kandel (2013). Liquidity cycles and make/take fees in electronic markets. *The Journal of Finance* 68(1), 299–341.
- Foucault, T., R. Kozhan, and W. W. Tham (2017). Toxic arbitrage. *The Review of Financial Studies* 30(4), 1053–1094.
- Foucault, T., A. Röell, and P. Sandås (2003). Market making with costly monitoring: An analysis of the soes controversy. *The Review of Financial Studies* 16(2), 345–384.
- Glosten, L. R. and P. R. Milgrom (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics* 14(1), 71–100.
- Goldstein, M. A. and K. A. Kavajecz (2004). Trading strategies during circuit breakers and extreme market movements. *Journal of Financial Markets* 7(3), 301–333.
- Hasbrouck, J. and G. Saar (2009). Technology and liquidity provision: The blurring of traditional definitions. *Journal of Financial Markets* 12(2), 143–172.
- Hendershott, T., C. M. Jones, and A. J. Menkveld (2011). Does algorithmic trading improve liquidity? *The Journal of Finance* 66(1), 1–33.
- Hendershott, T., S. Khan, and R. Riordan (2022). Option auctions. SSRN Working Paper. No. 4110516, Available at: <https://papers.ssrn.com/abstract=4110516>.
- Hendershott, T. and R. Riordan (2013). Algorithmic trading and the market for liquidity. *Journal of Financial and Quantitative Analysis* 48(4), 1001–1024.
- Hirschey, N. (2021). Do high-frequency traders anticipate buying and selling pressure? *Management Science* 67(6), 3321–3345.
- Hoffmann, P. (2014). A dynamic limit order market with fast and slow traders. *Journal of Financial Economics* 113(1), 156–169.
- Johnson, T. L. and E. C. So (2012). The option to stock volume ratio and future returns. *Journal of Financial Economics* 106(2), 262–286.
- Jones, C. M. (2013). What do we know about high-frequency trading? Columbia Business School Research Paper. No. 13-11, Available at SSRN: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2236201.
- Karlin, S. and H. M. Taylor (1975). *A First Course in Stochastic Processes* (2nd ed.). New York: Academic Press.
- Korajczyk, R. A. and D. Murphy (2019). High-frequency market making to large institutional trades. *The Review of Financial Studies* 32(3), 1034–1067.
- Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica* 53(6), 1315–1335.

- Li, S., X. Wang, and M. Ye (2021). Who provides liquidity, and when? *Journal of Financial Economics* 141(3), 968–980.
- Menkveld, A. J. (2013). High frequency trading and the new market makers. *Journal of Financial Markets* 16(4), 712–740.
- Menkveld, A. J. (2016). The economics of high-frequency trading: Taking stock. *Annual Review of Financial Economics* 8(1), 1–24.
- Muravyev, D. and N. D. Pearson (2020). Options trading costs are lower than you think. *The Review of Financial Studies* 33(11), 4973–5014.
- Ni, S. X., J. Pan, and A. M. Poteshman (2008). Volatility information trading in the option market. *The Journal of Finance* 63(3), 1059–1091.
- Ni, S. X., N. D. Pearson, A. M. Poteshman, and J. White (2021). Does option trading have a pervasive impact on underlying stock prices? *The Review of Financial Studies* 34(4), 1952–1986.
- Nimalendran, M., K. Rzaev, and S. Sagade (2024). High-frequency trading in the stock market and the costs of option market making. *Journal of Financial Economics* 152, 103310.
- O’Hara, M. (2015). High frequency market microstructure. *Journal of Financial Economics* 116(2), 257–270.
- Pan, J. and A. M. Poteshman (2006). The information in option volume for future stock prices. *The Review of Financial Studies* 19(3), 871–908.
- Shkilko, A. and K. Sokolov (2020). Every cloud has a silver lining: Fast trading, microwave connectivity, and trading costs. *The Journal of Finance* 75(6), 2899–2927.
- Xu, Z. and X. Cheng (2023). High-frequency anticipatory trading and its influences: Small informed trader vs. front-runner. SSRN Working Paper. No. 4430660, Available at: <https://papers.ssrn.com/abstract=4430660>.
- Yang, L. and H. Zhu (2020). Back-running: Seeking and hiding fundamental information in order flows. *The Review of Financial Studies* 33(4), 1484–1533.
- Yao, C. and M. Ye (2018). Why trading speed matters: A tale of queue rationing under price controls. *The Review of Financial Studies* 31(6), 2157–2183.

5 Figures

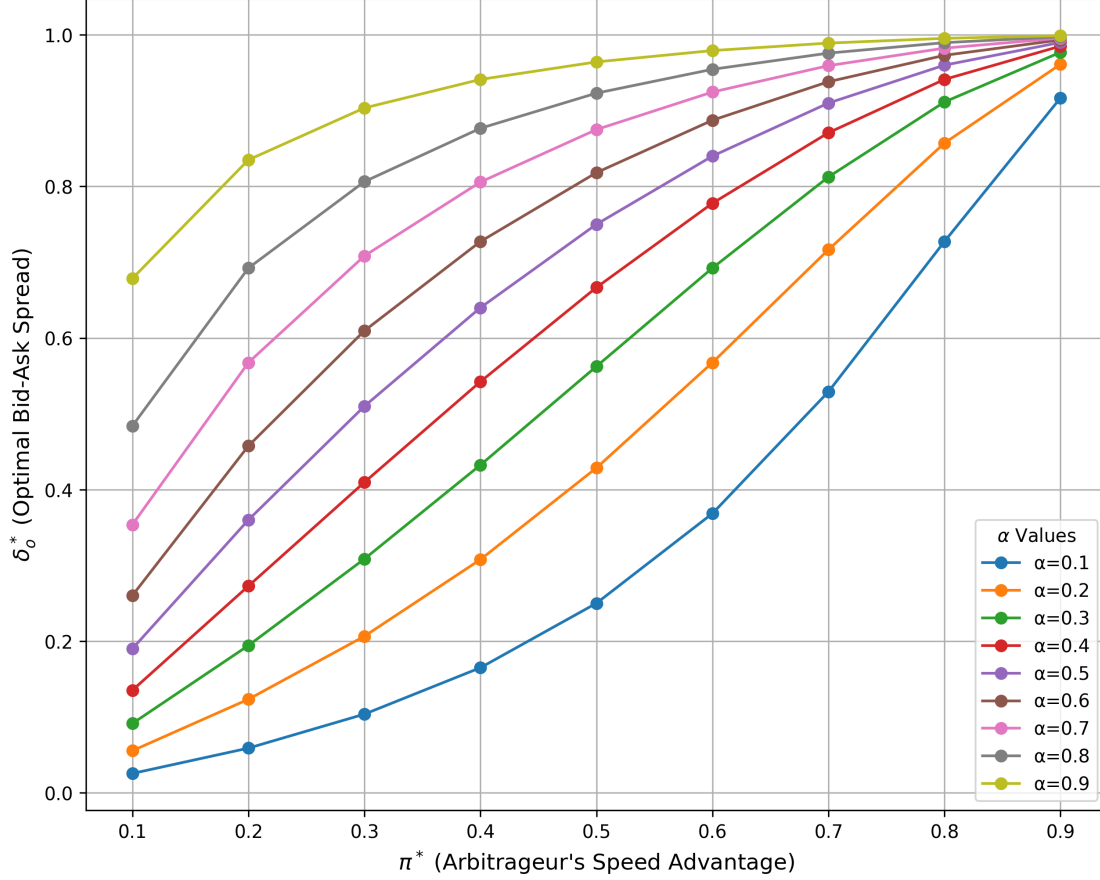


Figure 1
Optimal Spread as a Function of Speed Advantage

The figure illustrates the relationship between the market maker's optimal bid-ask spread (δ_{opt}^*) and the arbitrageur's speed advantage (π^*) under different levels of stock volatility (α). The estimates are generated from the equation (8). The spread adjustment follows a convex pattern for low α , where adverse selection risks are initially low, but increases sharply as π^* rises. For high α , the relationship transitions to concave, as the spread is already wide due to heightened volatility, reducing the marginal impact of π^* . The arbitrage speed has a stronger influence on spreads in low-volatility environments, while in high-volatility conditions, spread adjustments are dominated by stock price fluctuations.

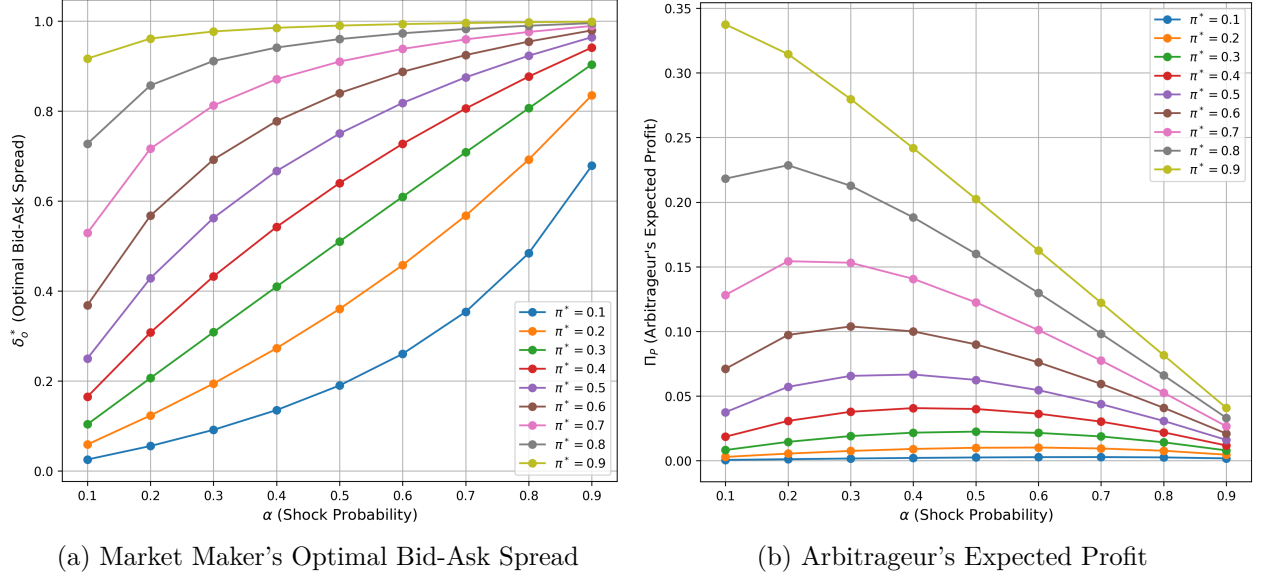
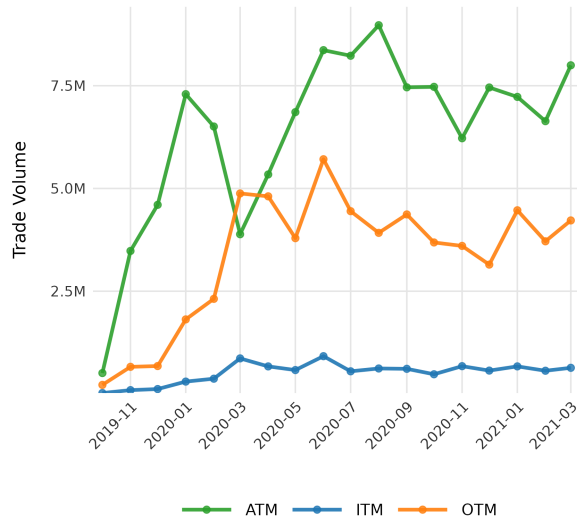
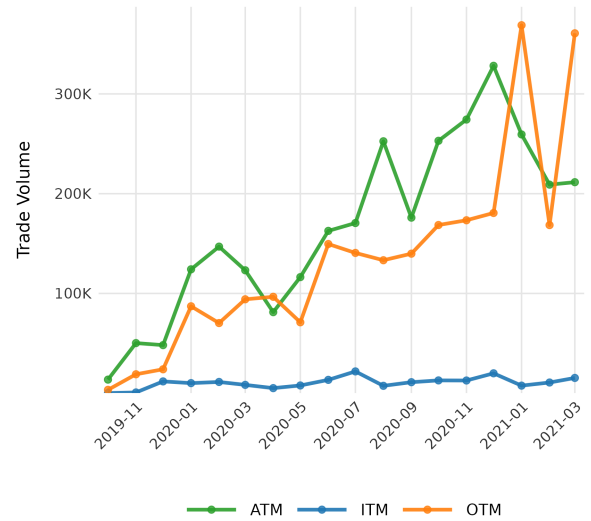


Figure 2
Optimal Spread and Arbitrageur's Profit

The figure on the left shows the market maker's optimal bid-ask spread (δ_{opt}^*) as a function of shock probability (α) for different levels of speed advantage (π^*). The figure on the right illustrates the arbitrageur's expected profit (Π_P) under the same conditions. The functional forms governing these relationships are given in equations (8) and (10). As α increases, market makers widen the spread to mitigate arbitrage risks, which reduces arbitrage profits. The arbitrageur's profit function exhibits a concave relationship with α , peaking at moderate values before declining due to defensive spread adjustments.



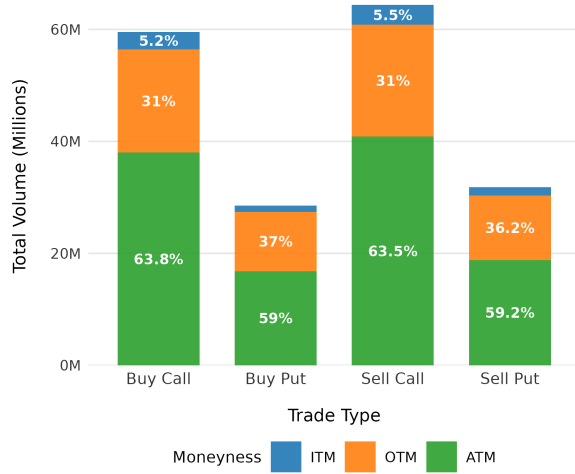
(a) Customer Volume



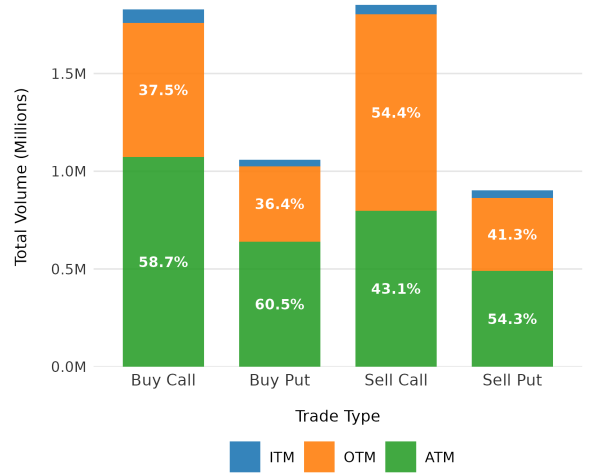
(b) Professional Volume

Figure 3
Options Trade Volume Trend

The figure presents the monthly trends in options trading volume by trader type and moneyness classification. The left figure displays customer trading volume patterns, while the right figure shows professional trader volume dynamics over the sample period. Trading volumes are aggregated by moneyness categories defined using a $\pm 5\%$ threshold relative to the underlying stock price. The sample comprises all CBOE options transactions on S&P 500 constituent stocks from October 2019 to March 2021.



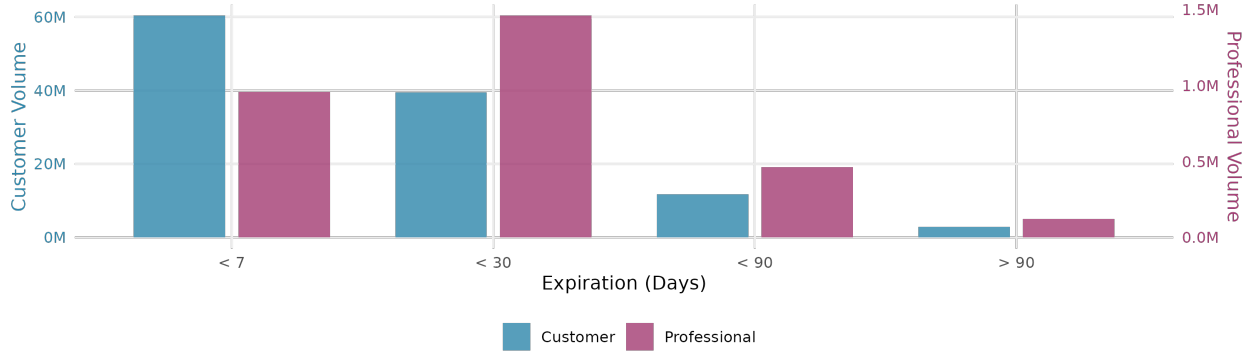
(a) Customer Volume



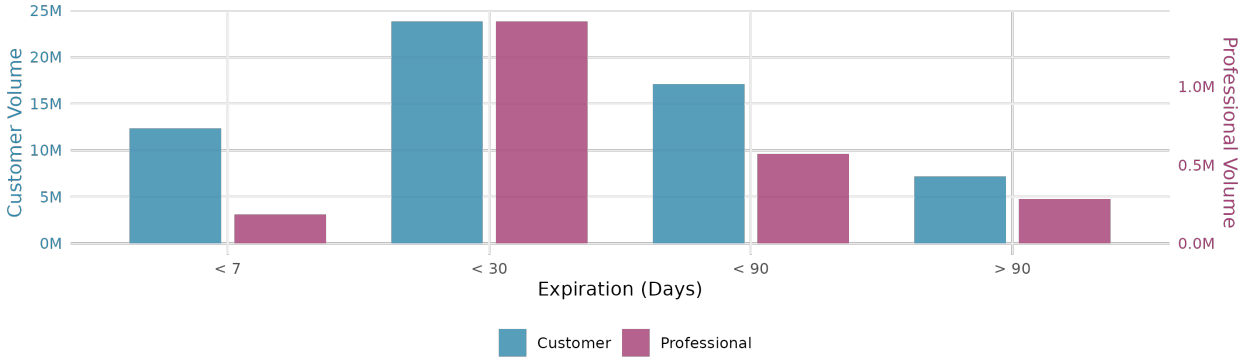
(b) Professional Volume

Figure 4
Option Trading Volume by Moneyiness

The figure presents the distribution of options trading volume across different moneyiness categories and agent types: customers (left) and professional customers (right). The stacked bars decompose total volume by moneyiness classification defined using a $\pm 5\%$ moneyiness threshold relative to the underlying stock price. Transaction types are categorized as buy calls, sell calls, buy puts, and sell puts. Percentage labels within each segment indicate the relative proportion of volume by moneyiness category within each transaction type.



(a) At-The-Money (ATM) Options



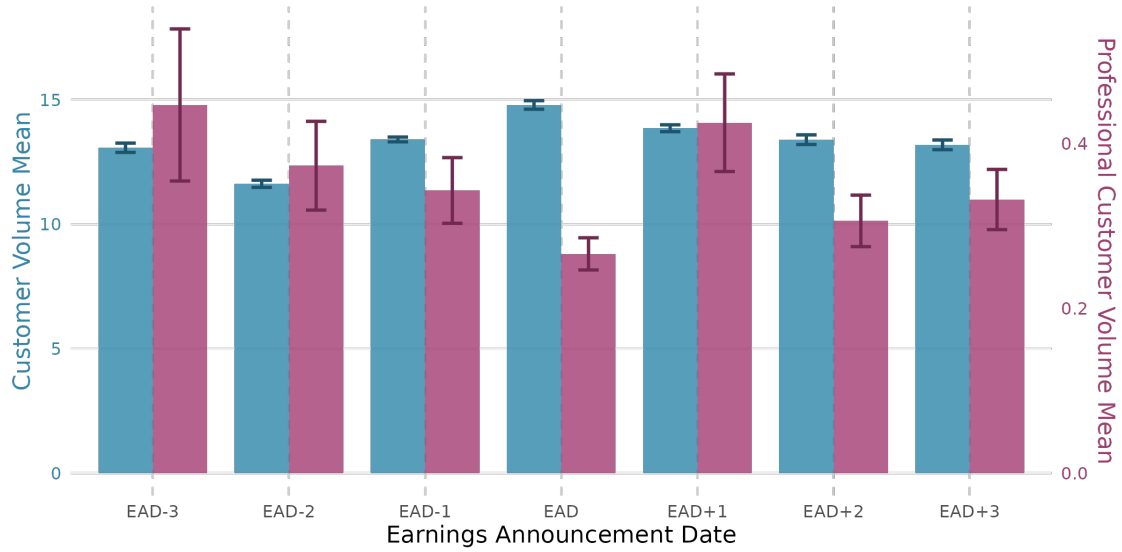
(b) Out-of-The-Money (OTM) Options



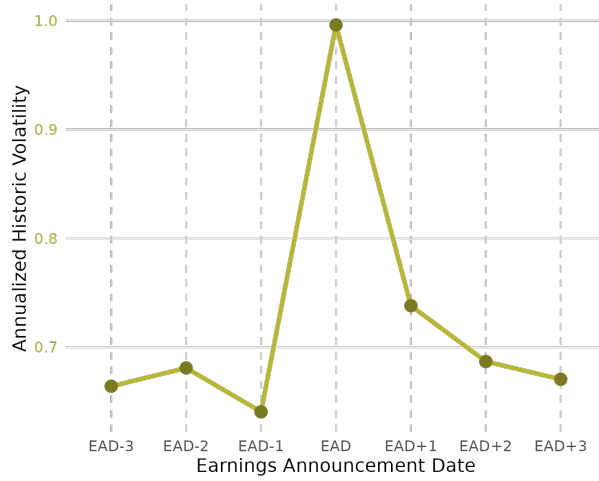
(c) In-The-Money (ITM) Options

Figure 5
Option Trading Volume by Expiration and Moneyness

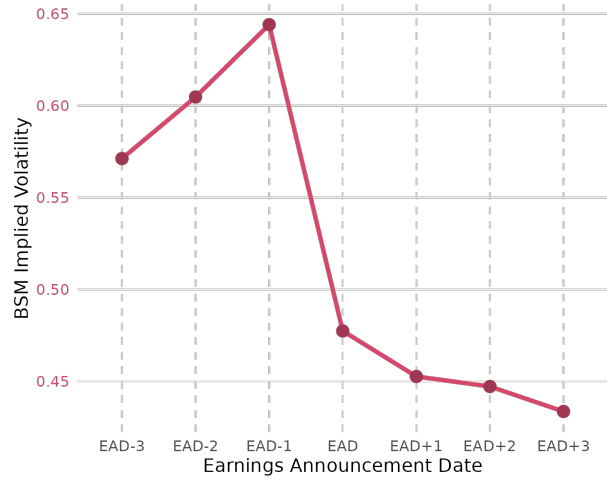
The plots compare option trading volumes between customer and professional customers across different moneyness categories and expiration periods. Each subplot uses dual y-axes to accommodate the scale differences between customer volumes (blue bars, left axis) and professional volumes (purple bars, right axis). Moneyness is determined by the stock-to-strike price ratio using a $\pm 5\%$ threshold. The time-to-expiration labels represent less than 7 days, less than 30 days, less than 90 days, and 90 days or more, respectively.



(a) Trade Volume



(b) Historic Volatility



(c) Implied Volatility

Figure 6
Options Trading Volume and Volatility around Earnings Announcement

The top figure presents options transaction volume from both customers and professional customers side-by-side, with professional customer volume scaled for comparison. Error bars represent the standard error of the mean. EAD represents the first business day affected by the earnings announcement. Historic volatility is calculated based on underlying returns in 10-minute bucket and annualized. Implied volatility is based on Black-Scholes-Merton model.

6 Tables

This page is intentionally left blank.

Table 1
Hypotheses Predictions

Hypothesis	Main Counterparty	Trade Behavior with Increased Volatility	Information Event	Return Predictability	Price Impact	Impact on Liquidity (Spread)
H1: Toxic Arbitrageurs	MMs	Increase	Increase	High	High	Increase
H2: Liquidity Suppliers (Non-toxic Arbitrage / Market-making)	Customers (Informed & Uninformed)	Proportional to Customers	Proportional to Customers	Comparable to MMs	Comparable to MMs	Decrease
H3: Cream-skimming (Pseudo Market-making)	Customers (Uninformed)	Proportional to Customers (Uninformed)	Decrease	Lower than MMs	Lower than MMs	Increase
H4: Informed Traders	MMs	Increase	Increase	High	High	Increase

Professional customers are exchange-identified high-frequency traders in the options market that are not market makers or dealers, who place orders at least 390 orders per day (i.e., one order per minute) on average. H1 proposes that PCs exploit speed advantages to trade against stale quotes and capitalize on short-term arbitrage opportunities, particularly during volatile periods. H2 suggests PCs act as alternative market makers, complementing traditional MMs by providing liquidity. H3 posits that PCs selectively provide liquidity to uninformed orders while avoiding informed trades, leaving MMs exposed to toxic order flow. H4 hypothesizes that PCs possess private information about future price movements or volatility and use this information advantage through strategic options trading.

Table 2
Descriptive statistics

Panel A: Options

Variable	N	Mean	SD	1%	25%	50%	75%	99%
Cust vol (contracts)	13,169,827	13.99	88.16	0.00	1.00	3.00	10.00	189.00
Prof vol (contracts)	13,169,827	0.43	33.46	0.00	0.00	0.00	0.00	3.00
MM vol (contracts)	13,169,827	13.93	91.54	1.00	1.00	3.00	10.00	189.00
Days to expire	13,169,827	33.41	74.92	0.00	4.00	14.00	32.00	392.00
Net Open interest	13,169,827	0.60	88.88	-99.00	-2.00	1.00	3.00	100.00
Quoted Spread (\$)	13,169,827	0.22	0.49	0.01	0.03	0.08	0.20	2.60
Effective Spread (%)	13,169,827	5.34	53.64	0.00	0.84	1.92	4.44	66.70
Quoted Spread (%)	13,169,827	11.70	23.47	0.73	2.57	4.80	10.40	141.00
Implied volatility (%)	13,099,943	51.0	35.0	15.00	30.00	40.00	59.00	196.00
DELTA	13,169,827	0.12	0.43	-0.96	-0.13	0.10	0.45	1.00
GAMMA	13,169,827	0.08	0.20	0.00	0.01	0.03	0.09	0.70
THETA	13,169,827	-0.19	0.52	-1.89	-0.19	-0.07	-0.02	0.00
VEGA	13,169,827	8.88	14.26	0.00	1.15	4.11	10.90	67.10
GammaExp	13,169,827	0.06	30.73	-10.80	-0.07	0.00	0.11	11.60
RebalCost	13,169,827	0.00	0.02	0.00	0.00	0.00	0.00	0.01

Panel B: Stocks

Variable	N	Mean	SD	1%	25%	50%	75%	99%
Stock return (10min, %)	12,994,508	0.00	0.59	-1.42	-0.17	0.00	0.17	1.42
Market return (10min, %)	12,992,307	0.00	0.21	-0.63	-0.07	0.00	0.08	0.62
Stock return (1day, %)	12,994,508	0.35	4.63	-10.90	-1.35	0.25	1.94	13.70
Market return (1day, %)	12,992,307	0.10	1.64	-5.06	-0.51	0.18	0.82	4.65
Volatility (10min, annual)	13,169,436	0.78	0.68	0.18	0.40	0.58	0.91	3.48
Mkt Volatility (10min, annual)	13,169,827	0.30	0.26	0.06	0.14	0.22	0.35	1.46
Quoted Spread (%)	13,169,827	0.04	0.04	0.01	0.02	0.03	0.05	0.20

The sample includes all CBOE options trades on S&P 500 stocks from October 2019 to March 2021. Customer, professional, and market maker volumes are represent the total number of options contracts transacted (buy/sell, open/close) at each 10-minute interval. Option Greeks are estimated using Black-Scholes model. Theta is reported on a per business day basis (252 days). Net OpenInterest is a measure of MMs' exposure, as defines in 3.1. *GammaExp* is a proxy for market maker exposure, calculated as NetOpenInterest multiplied by Gamma. Rebalancing cost is defined as the product of gamma, proportional stock spread, and the absolute value of open interest. Volatility and market volatility are measured as the realized standard deviation of underlying stock returns over each 10-minute interval.

Table 3
Option Spreads by Moneyness

Panel A: At the Money								
Variable	N	Mean	SD	1%	25%	50%	75%	99%
Effective Spread (%)	7,670,700	4.26	55.3	0.00	0.83	1.78	3.83	41.90
Quoted Spread (%)	7,670,700	8.82	16.9	0.72	2.36	4.24	8.41	82.40
Quoted Spread (\$)	7,670,700	0.20	0.41	0.01	0.03	0.08	0.20	2.09

Panel B: Out of the Money								
Variable	N	Mean	SD	1%	25%	50%	75%	99%
Effective Spread (%)	4,604,673	7.83	29.2	0.00	1.08	2.64	6.45	88.60
Quoted Spread (%)	4,604,673	17.8	32.0	0.94	3.45	6.92	16.40	200.00
Quoted Spread (\$)	4,604,673	0.18	0.45	0.01	0.02	0.06	0.16	2.25

Panel C: In the Money								
Variable	N	Mean	SD	1%	25%	50%	75%	99%
Effective Spread (%)	894,454	1.82	108.0	0.00	0.34	0.85	1.82	13.80
Quoted Spread (%)	894,454	5.01	8.23	0.47	1.67	2.88	5.27	37.00
Quoted Spread (\$)	894,454	0.60	0.94	0.01	0.11	0.25	0.62	4.40

Moneyness is defined by the stock-to-strike price ratio, with a $\pm 5\%$ threshold. The quoted spread (in dollars) is the difference between the best ask and best bid. The proportional bid-ask spread (in percent) is the quoted spread expressed as a percentage of the bid-ask midpoint. The effective spread (in percent) is twice the absolute difference between the trade price and the midpoint, expressed as a percentage of the midpoint.

Table 4
Correlation Coefficients

Panel A: Total Volumes

Variable	Cust Buy	Prof Buy	MM Buy	Cust Sell	Prof Sell	MM Sell
Cust Buy	1.00	0.00	0.03	0.05	0.36	0.41
Prof Buy		1.00	0.03	0.38	0.00	0.01
MM Buy			1.00	0.42	0.01	0.03
Cust Sell				1.00	0.00	0.03
Prof Sell					1.00	0.03
MM Sell						1.00

Panel B: Calls

Variable	Cust BuyCall	Cust SellCall	Prof BuyCall	Prof SellCall	MM BuyCall	MM SellCall
Cust BuyCall	1.00	0.06	0.00	0.39	0.04	0.48
Cust SellCall		1.00	0.38	0.00	0.43	0.05
Prof BuyCall			1.00	0.00	0.04	0.01
Prof SellCall				1.00	0.00	0.04
MM BuyCall					1.00	0.05
MM SellCall						1.00

Panel C: Puts

Variable	Cust BuyPut	Cust SellPut	Prof BuyPut	Prof SellPut	MM BuyPut	MM SellPut
Cust BuyPut	1.00	0.03	0.00	0.29	0.03	0.32
Cust SellPut		1.00	0.37	0.00	0.40	0.02
Prof BuyPut			1.00	0.00	0.02	0.03
Prof SellPut				1.00	0.02	0.02
MM BuyPut					1.00	0.02
MM SellPut						1.00

Panel A reports correlations for each trader type’s signed transaction volume in 10-minute intervals. Panels B and C report correlations for signed volumes of call and put options, respectively. Volume variables are labeled by trader type, trade direction, and option type. Cust, Prof, and MM denote customers, professional customers, and market makers. For example, “Cust BuyPut” refers to customers’ buy-side transaction volume in put options.

Table 5
Information Content of Customer Trading

Dependent Variables:	Concurrent (10min, %) (1)	Future (10min forward, %) (2)	Future (20min forward, %) (3)
Customer Buy Call (std)	33.90*** (6.422)	2.219 (1.976)	1.258 (1.260)
Customer Sell Call (std)	0.8354 (2.435)	-0.2175 (1.605)	-0.0171 (1.015)
Customer Buy Put (std)	-18.70*** (5.914)	8.330 (10.51)	1.063 (1.660)
Customer Sell Put (std)	-1.085 (1.571)	-1.602 (1.601)	3.585*** (1.226)
Controls	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes
Observations	12,555,376	11,991,692	11,556,284
R ²	0.01604	0.00278	0.00215

All dependent variables are underlying excess market returns ($r_{i,t} - r_{mkt,t}$) expressed in percentages and annualized from 10-minute returns. Concurrent return (model 1) refers to the same 10-minute interval as the options trading, while future returns (models 2-3) capture subsequent stock performance in following intervals. Control variables include lagged excess returns (20-minute and 10-minute past realized returns to control for stock momentum effects), realized volatility, market volatility, rebalancing cost, and gamma exposure. GammaExposure is defined as $NetOpenInterest_{j,t}$ multiplied by option gamma (Γ_t). Customer volumes are standardized. Two-way fixed effects include option-day and time fixed effects. Standard errors are two-way clustered by option-day and time and shown in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 6
Information Content of Customer Trading: 2SLS Results

Dependent Variables:	Concurrent (%) (10min)	Future (%) (10min forward)	Concurrent (%) (10min)	Future (%) (10min forward)
	PCs		MMs	
	(1)	(2)	(3)	(4)
Customer Buy Call (std)	7.004* (3.909)	1.432 (2.415)	69.23*** (14.40)	0.9928 (3.733)
Customer Sell Call (std)	-3.475 (3.001)	-3.357* (2.022)	6.339 (5.075)	-2.166 (2.472)
Customer Buy Put (std)	-4.358 (4.166)	-4.701 (4.596)	-27.50*** (7.622)	9.378 (7.457)
Customer Sell Put (std)	1.622 (1.201)	0.4611 (2.101)	-4.596 (3.720)	-2.718 (3.272)
Controls	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes
F-test (1st stage), Buy Call	191,978.3	172,212.5	1,050,022.8	965,332.3
F-test (1st stage), Sell Call	191,373.3	177,301.8	658,789.9	594,762.9
F-test (1st stage), Buy Put	107,675.9	98,813.0	330,371.5	302,954.6
F-test (1st stage), Sell Put	177,591.1	112,873.0	633,182.9	621,782.3
Observations	12,555,376	11,991,692	12,555,376	11,991,692
R ²	0.01600	0.00278	0.01599	0.00278

The table reports coefficient estimates from the fixed effects model in equation 13, estimated at the 10-minute frequency. All dependent variables are underlying excess market returns ($r_{i,t} - r_{mkt,t}$) expressed in percentages and annualized from 10-minute returns. Concurrent return refers to the same 10-minute interval as the options trading, while future return measures subsequent stock performance. All four signed customer volumes are fitted values from equation 12. Models 1-2 use professional customer volumes as instruments, while models 3-4 use market maker volumes as instruments. Control variables include lagged excess returns (20-minute and 10-minute past realized returns to control for momentum effects), realized volatility, market volatility, rebalancing cost, and gamma exposure. All trader volumes are standardized. Two-way fixed effects include option-day and time fixed effects. Standard errors are two-way clustered by option-day and time and shown in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table 7
Return Predictability, Professional Customers

Dependent Variable: Moneyness Model:	OLS				2SLS			
	Concurrent 10-Minute Excess Return (% Annualized)				Concurrent 10-Minute Excess Return (% Annualized)			
	Full Sample (1)	ATM (2)	OTM (3)	ITM (4)	Full Sample (5)	ATM (6)	OTM (7)	ITM (8)
Buy Call Volume (Standardized)	-1.417 (1.053)	-2.270* (1.339)	-0.3565 (1.507)	-7.768 (9.211)	-1.533 (4.675)	2.807 (7.934)	1.679 (3.882)	-108.9 (74.82)
Sell Call Volume (Standardized)	2.062 (1.328)	2.381 (1.685)	0.7593 (1.703)	10.87 (10.61)	17.50*** (6.302)	26.21*** (7.754)	7.883 (6.030)	28.94 (35.82)
Buy Put Volume (Standardized)	0.1454 (0.5249)	0.0211 (0.5110)	0.6201 (1.356)	-5.750 (9.384)	2.870 (2.821)	1.848 (1.809)	7.301 (7.723)	200.2 (199.7)
Sell Put Volume (Standardized)	-0.9247 (1.126)	-0.5749 (0.9946)	-1.489 (2.075)	2.173 (4.268)	-30.94 (19.12)	-21.79 (13.49)	-8.931*** (2.791)	-751.2 (868.9)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F-test (1st Stage), Buy Call					232,868.4	88,035.1	126,108.1	5,242.2
F-test (1st Stage), Sell Call					240,646.7	77,750.5	111,495.8	10,460.3
F-test (1st Stage), Buy Put					200,999.5	220,023.7	42,413.3	915.07
F-test (1st Stage), Sell Put					128,959.0	63,455.5	56,447.9	8,540.7
Observations	12,555,376	7,337,832	4,384,249	833,295	12,555,376	7,337,832	4,384,249	833,295
R ²	0.01598	0.00600	0.02224	0.02619	0.01594	0.00594	0.02223	0.02336

The dependent variable is the annualized concurrent 10-minute excess return of the underlying stock. In 2SLS models (5-8), professional customer volumes are fitted with customer trades, consistent with equation 14. Control variables include 20-minute and 10-minute lagged realized returns (to capture stock momentum), realized volatility, market volatility, rebalancing cost, gamma exposure, days to expiration, inverse option mid-quote, implied volatility, stock bid-ask spread, absolute delta, and vega. Two-way fixed effects include option-day and time. Moneyness is based on stock-to-strike price ratio with $\pm 5\%$ threshold. Standard errors are two-way clustered by option-day and time and are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 8
Price Impact of Customer Trading: 2SLS Results

Dependent Variable:	Absolute Return (%) (10min concurrent)		
	Baseline (1)	PCs (2)	MMs (3)
Customer Buy Call (std)	13.19*** (2.751)	1.357 (1.165)	30.60*** (7.094)
Customer Sell Call (std)	9.678*** (2.369)	0.7851 (1.412)	23.08*** (5.961)
Customer Buy Put (std)	13.14** (5.240)	-2.084 (2.280)	21.53*** (4.790)
Customer Sell Put (std)	2.709* (1.560)	-3.953* (2.365)	13.29*** (3.607)
Controls	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes
F-test (1st stage), Buy Call		197,135.3	1,148,242.6
F-test (1st stage), Sell Call		192,207.3	740,318.6
F-test (1st stage), Buy Put		101,135.8	359,825.4
F-test (1st stage), Sell Put		179,832.6	683,157.8
Observations	12,994,269	12,994,269	12,994,269
R ²	0.26965	0.26963	0.26963

The table reports coefficient estimates for the price impact of customer options trading through two counterparty types. The dependent variable is the absolute concurrent 10-minute stock return (in percent, annualized). Model 1 presents baseline OLS results. Models 2–3 present 2SLS estimates: in Model 2, all four signed customer volumes are instrumented with professional customer volumes in the first stage; in Model 3, they are instrumented with market maker volumes. The fitted values from these first-stage regressions are then used in the main specification. Control variables include underlying realized volatility, market volatility, and absolute gamma exposure. All trader volumes are standardized. Two-way fixed effects include option-day and time fixed effects. Standard errors are two-way clustered by option-day and time and shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 9
Logistic Regression on Professional Trading Occurrence

Dependent Variable:	Professional Trade Indicator			
	Full Sample (1)	ATM (2)	OTM (3)	ITM (4)
Implied Volatility (std.)	-0.2031*** (0.0252)	-0.2116*** (0.0338)	-0.0284 (0.0294)	0.1463*** (0.0429)
Historic Volatility (std.)	0.1095*** (0.0197)	0.1927*** (0.0204)	-0.0038 (0.0244)	-0.1777*** (0.0338)
EAD (Day Before)	-0.6197*** (0.0574)	-0.5716*** (0.0563)	-0.8098*** (0.0868)	0.1321 (0.1234)
EAD	-0.6343*** (0.0596)	-0.5012*** (0.0593)	-0.9059*** (0.0915)	-0.7760*** (0.1273)
EAD (Day After)	-0.0386 (0.0402)	0.0001 (0.0450)	-0.1079** (0.0502)	-0.1432 (0.1074)
Medium Cust Order (>100 contracts)	0.6781*** (0.0769)	0.6438*** (0.0850)	0.5961*** (0.0433)	0.2553 (0.1796)
Large Cust Order (>200 contracts)	1.371*** (0.0838)	1.171*** (0.0749)	1.620*** (0.1315)	1.944*** (0.1896)
Controls	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes
Observations	13,098,533	7,647,539	4,596,638	844,366

The dependent variable is a dummy indicating the presence of any professional trade within a 10-minute interval. Historic volatility is measured from the stock's realized volatility over the preceding 10-minute interval, annualized and standardized. Control variables include delta, vega, rebalancing cost, relative stock and option spreads, market volatility, and days to expiration. *Medium Cust Order* is a dummy equal to 1 if any single customer order exceeds 100 contracts within the 10-minute interval; *Large Cust Order* is a dummy equal to 1 if any single order exceeds 200 contracts. Earnings Announcement (EAD) is a dummy for the day of the announcement. Two-way fixed effects include both option-day and time. Moneyness is defined by the stock-to-strike price ratio with a $\pm 5\%$ threshold. Implied volatility is estimated using the Black-Scholes model and standardized. Standard errors are two-way clustered by option-day and time, shown in parentheses (** $p < 0.01$, * $p < 0.05$, $p < 0.1$).

Table 10
Poisson Regression on Trading Activity

Panel A: PCs					Panel B: Customers				
Variable	Full Sample	ATM	OTM	ITM	Variable	Full Sample	ATM	OTM	ITM
Implied Volatility (std.)	-0.2587** (0.1226)	-0.1805 (0.1345)	-0.3768** (0.1640)	-0.0140 (0.0530)	Implied Volatility (std.)	0.0663*** (0.0102)	0.1055*** (0.0123)	0.0725*** (0.0154)	0.0233*** (0.0089)
Historic Volatility (std.)	0.0340** (0.0160)	-0.0113 (0.0474)	0.0713 (0.0767)	0.0323 (0.0770)	Histroic Volatility (std.)	0.0749*** (0.0096)	0.1472*** (0.0140)	0.0668*** (0.0086)	0.0392*** (0.0072)
EAD (Before)	0.2693 (0.2809)	0.5157** (0.2574)	0.2073 (0.3727)	0.2816 (0.8800)	EAD (Before)	0.0627*** (0.0222)	0.0145 (0.0289)	0.1061*** (0.0287)	-0.0873* (0.0520)
EAD (Day)	-0.2615 (0.2323)	-0.3908 (0.2601)	-0.2998 (0.2849)	0.0314 (0.2814)	EAD (Day)	0.0748*** (0.0161)	0.0977*** (0.0228)	0.0573** (0.0274)	0.0231 (0.0353)
EAD (After)	-0.0155 (0.2448)	-0.2798 (0.2213)	0.3106 (0.4044)	0.0325 (0.2117)	EAD (After)	0.0388** (0.0163)	0.0553*** (0.0146)	0.0245 (0.0275)	-0.0381 (0.0397)
Controls	Yes	Yes	Yes	Yes	Controls	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes	Fixed Effects	Yes	Yes	Yes	Yes
Observations	13,098,533	7,647,539	4,596,638	844,366	Observations	13,099,558	7,648,887	4,599,466	851,205

Panel A reports Poisson regression results for PC trades in a 10-minute interval, weighted by predicted probabilities from a logistic regression model of professional trade occurrence. Panel B reports Poisson regression results for the number of customer trades in a 10-minute interval. We compare PC and customer since MMs take passive roles in trade initiation. Historic volatility is defined as the stock's realized volatility over the preceding 10-minute interval, annualized and standardized. Implied volatility is estimated using the Black–Scholes model and standardized. Control variables include medium (a dummy equal to 1 if any single customer order exceeds 100 contracts within the 10-minute interval), large (a dummy equal to 1 if any single order exceeds 200 contracts within the 10-minute interval), delta, vega, rebalancing cost, relative stock and option spreads, market volatility, and days to expiration. Earnings Announcement (EAD) refers to the day of the announcement; EAD (Before) is the day prior; and EAD (After) is the following day. Two-way fixed effects include both option-day and time. Two-way fixed effects include option-day and time. Moneyness is based on stock-to-strike price ratio with a $\pm 5\%$ threshold. Standard errors are two-way clustered by option-day and time, shown in parentheses (** $p < 0.01$, * $p < 0.05$, $p < 0.1$).

Table 11
Effective Spreads by Moneyness

Dependent Variable Moneyness	Effective Spread (%)											
	ATM				OTM				ITM			
	OLS	Two-Stage	OLS	Two-Stage	OLS	Two-Stage	OLS	Two-Stage	OLS	Two-Stage	OLS	Two-Stage
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Market Maker Volume (std)	-0.1463* (0.0773)	-0.8441*** (0.1260)	-0.1372* (0.0735)	-0.6677*** (0.1251)	-0.0003 (0.0684)	-0.2330* (0.1277)	0.0002 (0.0708)	-0.0688 (0.1245)	-0.0108** (0.0052)	-0.0573** (0.0224)	-0.0112** (0.0045)	-0.0127 (0.0192)
Professional Volume (std)	0.0203*** (0.0057)	0.5078*** (0.1839)			0.0414*** (0.0086)	0.2105*** (0.0699)			0.0127*** (0.0036)	0.1665* (0.0995)		
MM Volume $\times$ Small Prof			-0.3564*** (0.0503)	0.0823 (0.0707)			-0.1409** (0.0565)	-0.0966 (0.0777)			-0.0282*** (0.0081)	-0.0272* (0.0145)
MM Volume $\times$ Medium Prof			0.0955 (0.0666)	0.0885 (0.1038)			0.1899* (0.1102)	0.2009* (0.1101)			-0.2662*** (0.0959)	-0.2722** (0.1230)
MM Volume $\times$ Large Prof			0.2662*** (0.0663)	0.5471*** (0.1154)			0.2998*** (0.0953)	0.3380*** (0.0986)			-0.0406*** (0.0117)	-0.0401*** (0.0124)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F-test (1st stage), MM Volume		139,306.2		122,521.6		259,632.6		243,411.4		26,503.1		23,574.5
F-test (1st stage), Prof Volume		109,309.4				101,119.2				7,027.3		
Observations	7,648,887	7,648,887	7,648,887	7,648,887	4,599,466	4,599,466	4,599,466	4,599,466	851,206	851,206	851,206	851,206
R ²	0.01133	0.01111	0.01133	0.01124	0.11309	0.11302	0.11309	0.11309	0.14306	0.14227	0.14307	0.14307

All volume variables are standardized. Control variables include days to expiration, inverse option mid-quote, historic volatility, market volatility, implied volatility, stock bid-ask spread, absolute delta, rebalancing cost, and vega. *Small Prof* indicates professional volume > 0, *Medium Prof* indicates volume > 100 contracts, *Large Prof* indicates volume > 200 contracts. Models (1), (3), (5), (7), (9), and (11) are OLS regressions. Models (2), (4), (6), (8), (10), and (12) are two-stage regressions where total volume of market maker and professional customers are fitted by customer volumes from equation 14. The fitted values then used in the main regression. Moneyness classification is based on stock-to-strike price ratio with $\pm 5\%$ threshold. Two-way fixed effects include option-day and time fixed effects and standard errors are also two-way clustered and shown in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

7 Internet Appendix

This page intentionally left blank.

Expected Value and Variance for the Bernoulli Process

Shocks occur according to a Bernoulli process with parameter α . When a shock occurs, there is equal probability of $+\frac{\sigma_s}{2}$ or $-\frac{\sigma_s}{2}$.

Expected Value

The expected value of $\tilde{\nu}_s$ is:

$$\mathbb{E}[\tilde{\nu}_s] = \mathbb{E}[\tilde{\nu}_s \mid \text{Shock}] \cdot \alpha + \mathbb{E}[\tilde{\nu}_s \mid \text{No Shock}] \cdot (1 - \alpha)$$

When a shock occurs:

$$\mathbb{E}[\tilde{\nu}_s \mid \text{Shock}] = \left(\mu_s + \frac{\sigma_s}{2}\right) \cdot 0.5 + \left(\mu_s - \frac{\sigma_s}{2}\right) \cdot 0.5 = \mu_s$$

When no shock occurs: $\mathbb{E}[\tilde{\nu}_s \mid \text{No Shock}] = \mu_s$.

Thus: $\mathbb{E}[\tilde{\nu}_s] = \mu_s \cdot \alpha + \mu_s \cdot (1 - \alpha) = \mu_s$.

Variance

The variance is $\text{Var}(\tilde{\nu}_s) = \mathbb{E}[\tilde{\nu}_s^2] - \mu_s^2$.

When a shock occurs:

$$\mathbb{E}[\tilde{\nu}_s^2 \mid \text{Shock}] = \left(\mu_s + \frac{\sigma_s}{2}\right)^2 \cdot 0.5 + \left(\mu_s - \frac{\sigma_s}{2}\right)^2 \cdot 0.5 = \mu_s^2 + \frac{\sigma_s^2}{4}$$

When no shock occurs: $\mathbb{E}[\tilde{\nu}_s^2 \mid \text{No Shock}] = \mu_s^2$.

Using the law of total expectation:

$$\mathbb{E}[\tilde{\nu}_s^2] = \left(\mu_s^2 + \frac{\sigma_s^2}{4}\right) \cdot \alpha + \mu_s^2 \cdot (1 - \alpha) = \mu_s^2 + \alpha \cdot \frac{\sigma_s^2}{4}$$

Therefore:

$$\text{Var}[\tilde{\nu}_s] = \alpha \cdot \frac{\sigma_s^2}{4}$$

Market Maker's Profit

We start with the market maker's profit function:

$$\Pi_M = -\alpha \pi^* \left(\frac{\sigma_{opt}}{2} - \frac{\delta_{opt}^*}{2} \right) + (1 - \alpha)(1 - \pi^*) \frac{\delta_{opt}^*}{2} - \frac{c_M}{2} \lambda_M^*. \quad (17)$$

Factor out $\frac{1}{2}$ from the first two terms:

$$\Pi_M = \frac{1}{2} \left[-\alpha \pi^* (\sigma_{opt} - \delta_{opt}^*) + (1 - \alpha)(1 - \pi^*) \delta_{opt}^* \right] - \frac{c_M}{2} \lambda_M^*. \quad (1)$$

In equilibrium, substituting λ_M^* and $\pi^* = c_M / (c_M + c_P)$

$$-\frac{c_M}{2} \lambda_M^* = -\frac{1}{2} \pi^* (1 - \pi^*) \left[\alpha (\sigma_{opt} - \delta_{opt}^*) + (1 - \alpha) \delta_{opt}^* \right]$$

Hence, the profit function becomes

$$\Pi_M = \frac{1}{2} \left[-\alpha \pi^* (\sigma_{opt} - \delta_{opt}^*) + (1 - \alpha)(1 - \pi^*) \delta_{opt}^* \right] - \frac{1}{2} \pi^* (1 - \pi^*) \left[\alpha (\sigma_{opt} - \delta_{opt}^*) + (1 - \alpha) \delta_{opt}^* \right] \quad (18)$$

$$= \frac{1}{2} \left[(1 - \alpha)(1 - \pi^*) \delta_{opt}^* - \alpha \pi^* (\sigma_{opt} - \delta_{opt}^*) - \pi^* (1 - \pi^*) \left(\alpha (\sigma_{opt} - \delta_{opt}^*) + (1 - \alpha) \delta_{opt}^* \right) \right]$$

Comparative Statics

Consider the equilibrium option spread given by

$$\delta_{\text{opt}}^* = \frac{\alpha \pi^*(2 - \pi^*)}{\pi^*(2 - \pi^*)(2\alpha - 1) + (1 - \alpha)} \sigma_{\text{opt}}, \quad (19)$$

and we set

$$\sigma_{\text{opt}} = 0.5.$$

For notational convenience, define

$$N(\alpha, \pi^*) = \alpha \pi^*(2 - \pi^*) \quad \text{and} \quad D(\alpha, \pi^*) = \pi^*(2 - \pi^*)(2\alpha - 1) + (1 - \alpha). \quad (20)$$

Thus, we can write

$$\delta_{\text{opt}}^* = s f(\alpha, \pi^*), \quad (21)$$

with

$$s = \sigma_{\text{opt}} = 0.5 \quad \text{and} \quad f(\alpha, \pi^*) = \frac{N(\alpha, \pi^*)}{D(\alpha, \pi^*)}.$$

These derivations are obtained under the assumption that $\sigma_{\text{opt}} = 0.5$ (i.e., $s = 0.5$), and the constant s does not affect the sign of the derivatives.

Partial Derivatives with Respect to α

First, note that

$$N(\alpha, \pi^*) = \alpha \pi^*(2 - \pi^*) \quad \Rightarrow \quad N_{\alpha}(\alpha, \pi^*) = \pi^*(2 - \pi^*). \quad (22)$$

Also,

$$D(\alpha, \pi^*) = \pi^*(2 - \pi^*)(2\alpha - 1) + (1 - \alpha), \quad (23)$$

so that

$$D_\alpha(\alpha, \pi^*) = \frac{\partial}{\partial \alpha} \left[\pi^*(2 - \pi^*)(2\alpha - 1) + (1 - \alpha) \right] = 2\pi^*(2 - \pi^*) - 1. \quad (24)$$

Using the quotient rule,

$$\frac{\partial f}{\partial \alpha} = \frac{N_\alpha D - N D_\alpha}{D^2} = \frac{\pi^*(2 - \pi^*) D(\alpha, \pi^*) - \alpha \pi^*(2 - \pi^*) [2\pi^*(2 - \pi^*) - 1]}{\left[D(\alpha, \pi^*) \right]^2}. \quad (25)$$

Factor out $\pi^*(2 - \pi^*)$:

$$\frac{\partial f}{\partial \alpha} = \frac{\pi^*(2 - \pi^*) \left[D(\alpha, \pi^*) - \alpha (2\pi^*(2 - \pi^*) - 1) \right]}{\left[D(\alpha, \pi^*) \right]^2}. \quad (26)$$

Substitute

$$D(\alpha, \pi^*) = \pi^*(2 - \pi^*)(2\alpha - 1) + (1 - \alpha),$$

to simplify the numerator:

$$D(\alpha, \pi^*) - \alpha (2\pi^*(2 - \pi^*) - 1) = \pi^*(2 - \pi^*)(2\alpha - 1) + (1 - \alpha) - \alpha (2\pi^*(2 - \pi^*) - 1).$$

Grouping the terms yields

$$D(\alpha, \pi^*) - \alpha (2\pi^*(2 - \pi^*) - 1) = 1 - \pi^*(2 - \pi^*).$$

Thus, the first partial derivative with respect to α is

$$\frac{\partial f}{\partial \alpha} = \frac{\pi^*(2 - \pi^*) \left[1 - \pi^*(2 - \pi^*) \right]}{\left[D(\alpha, \pi^*) \right]^2}, \quad (27)$$

and therefore,

$$\frac{\partial \delta_{\text{opt}}^*}{\partial \alpha} = s \frac{\pi^*(2 - \pi^*) \left[1 - \pi^*(2 - \pi^*) \right]}{\left[D(\alpha, \pi^*) \right]^2}. \quad (28)$$

Since $\pi^*(2 - \pi^*) > 0$ and $1 - \pi^*(2 - \pi^*) \geq 0$ (for $\pi^* < 1$), this derivative is positive.

Differentiating once more with respect to α gives the second derivative:

$$\frac{\partial^2 f}{\partial \alpha^2} = - \frac{2 \pi^*(2 - \pi^*)(1 - \pi^*(2 - \pi^*))(2\pi^*(2 - \pi^*) - 1)}{\left[D(\alpha, \pi^*) \right]^3}, \quad (29)$$

so that

$$\frac{\partial^2 \delta_{\text{opt}}^*}{\partial \alpha^2} = s \frac{\partial^2 f}{\partial \alpha^2} = -2s \frac{\pi^*(2 - \pi^*)(1 - \pi^*(2 - \pi^*))(2\pi^*(2 - \pi^*) - 1)}{\left[D(\alpha, \pi^*) \right]^3}. \quad (30)$$

The sign of $\frac{\partial^2 \delta_{\text{opt}}^*}{\partial \alpha^2}$ depends on the factor $[2\pi^*(2 - \pi^*) - 1]$. In particular, if

$$2\pi^*(2 - \pi^*) < 1 \quad (\text{i.e., for sufficiently low } \pi^*),$$

then $2\pi^*(2 - \pi^*) - 1 < 0$ and the negative sign makes the second derivative positive; if $2\pi^*(2 - \pi^*) > 1$ the second derivative is negative.

Partial Derivatives with Respect to π^*

For fixed α , the derivative of $f(\alpha, \pi^*)$ with respect to π^* is given by

$$\frac{\partial f}{\partial \pi^*} = \frac{2\alpha(1 - \alpha)(1 - \pi^*)}{\left[D(\alpha, \pi^*) \right]^2}, \quad (31)$$

so that

$$\frac{\partial \delta_{\text{opt}}^*}{\partial \pi^*} = s \frac{2\alpha(1 - \alpha)(1 - \pi^*)}{\left[D(\alpha, \pi^*) \right]^2}. \quad (32)$$

Since $\alpha(1 - \alpha) > 0$ (for $\alpha \in (0, 1)$) and $1 - \pi^* > 0$ for $\pi^* < 1$, this derivative is positive.

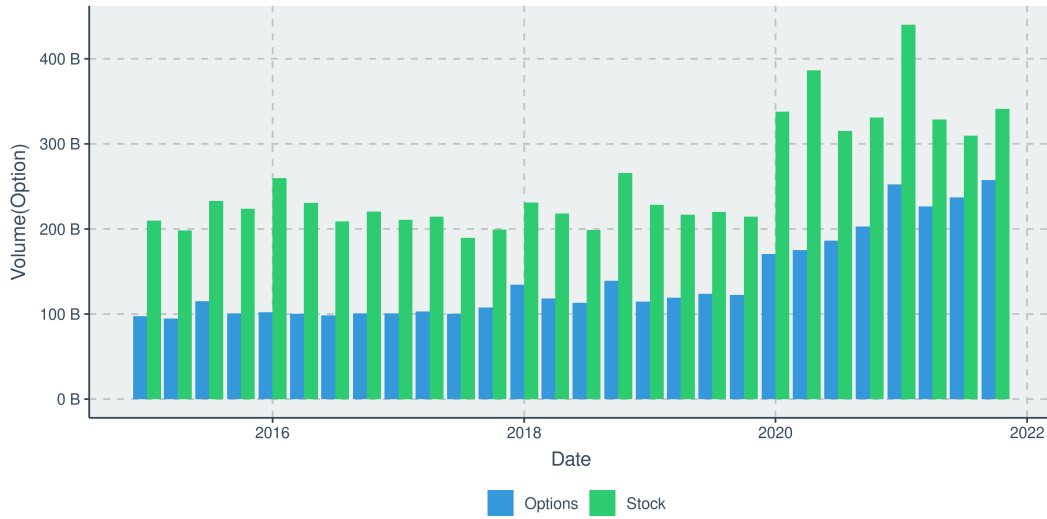
Differentiating again with respect to π^* gives the second derivative:

$$\frac{\partial^2 f}{\partial(\pi^*)^2} = -\frac{2\alpha(1-\alpha)}{\left[D(\alpha, \pi^*)\right]^2} - \frac{4\alpha(1-\alpha)(2\alpha-1)(1-\pi^*)^2}{\left[D(\alpha, \pi^*)\right]^3}, \quad (33)$$

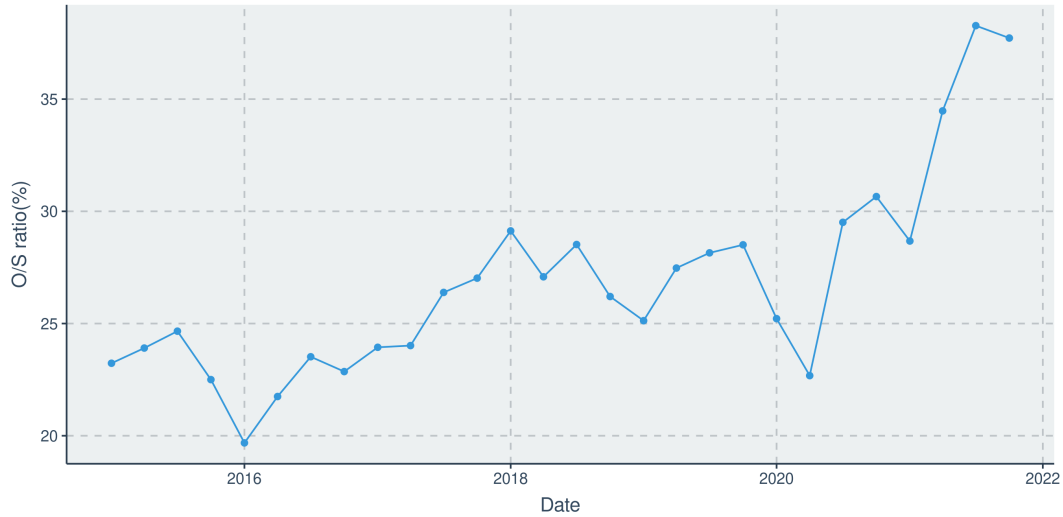
so that

$$\frac{\partial^2 \delta_{\text{opt}}^*}{\partial(\pi^*)^2} = s \left[-\frac{2\alpha(1-\alpha)}{\left[D(\alpha, \pi^*)\right]^2} - \frac{4\alpha(1-\alpha)(2\alpha-1)(1-\pi^*)^2}{\left[D(\alpha, \pi^*)\right]^3} \right]. \quad (34)$$

The overall sign depends on the level of α : for $\alpha < 0.5$ (so that $2\alpha - 1 < 0$) the second term may be positive and dominate (indicating convexity in π^*), while for $\alpha > 0.5$ the expression is negative (indicating concavity).



(a) Quarterly Option and Stock Volume Traded in U.S.



(b) Options to Stock ratio trend

Figure A1
Option and Stock Trade Volume Trend

The figure above shows the quarterly volume trend from Q1 2015 to Q4 2021 for all U.S. exchanges and NASDAQ-listed equities and equity options, as reported by OptionMetrics. Options volume is measured as the number of contracts traded multiplied by the number of underlying shares ($\times 100$), since each option contract represents 100 shares of the underlying equity. The figure below shows the quarterly trend in the options-to-stocks (O/S) ratio, expressed as a percentage.

Table A1
Information Content of Customer Trading by Moneyness

Dependent Variables:	ATM			OTM			ITM		
	Concurrent (%)	Future (%)	Future (%)	Concurrent (%)	Future (%)	Future (%)	Concurrent (%)	Future (%)	Future (%)
	(10min)	(10min forward)	(10min holding)	(10min)	(10min forward)	(10min holding)	(10min)	(10min forward)	(10min holding)
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Customer Buy Call (std)	56.17*** (9.952)	1.216 (2.537)	60.29*** (10.75)	12.81*** (2.979)	2.333 (1.697)	15.96*** (3.065)	10.00 (20.77)	34.31 (38.14)	28.57 (18.56)
Customer Sell Call (std)	3.796 (3.673)	-1.484 (1.969)	2.026 (5.015)	-0.9736 (2.462)	0.9483 (1.915)	0.0847 (3.907)	-12.96 (11.26)	-6.484 (12.95)	-16.79 (13.87)
Customer Buy Put (std)	-21.71*** (5.545)	0.2510 (1.843)	-22.60*** (6.174)	-6.250*** (0.9547)	-5.307 (4.145)	-12.20*** (4.181)	-188.9 (166.8)	328.9 (317.1)	-31.78 (28.31)
Customer Sell Put (std)	-5.202* (2.784)	-0.0974 (1.639)	-4.859 (3.701)	3.932 (2.462)	-1.706 (1.774)	2.366 (3.038)	1.423 (11.16)	-9.876 (15.60)	-4.297 (8.191)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,337,832	7,015,567	7,015,567	4,384,249	4,180,466	4,180,466	833,295	795,659	795,659
R ²	0.00621	0.00203	0.00868	0.02224	0.00325	0.02758	0.02647	0.00803	0.03681

All dependent variables are underlying excess market returns (in percentage terms), annualized from 10-minute interval returns. Concurrent stock returns correspond to the same 10-minute intervals as options trading, while future returns measure subsequent stock performance. Holding returns are defined as cumulative holding-period returns. Control variables include lagged excess returns (20-minute and 10-minute prior realized returns to capture momentum effects), realized volatility, market volatility, rebalancing costs, and gamma exposure. Customer volumes are standardized. Two-way fixed effects include option-day and time fixed effects. Moneyness classification is based on stock-to-strike price ratio with $\pm 5\%$ threshold. Standard errors are two-way clustered by option-day and time and shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A2
Information Content of Customer Trading by Moneyness: 2SLS on PCs

Dependent Variables:	ATM		OTM		ITM	
	Concurrent (%) (10min) (1)	Future (%) (10min forward) (2)	Concurrent (%) (10min) (3)	Future (%) (10min forward) (4)	Concurrent (%) (10min) (5)	Future (%) (10min forward) (6)
Model:						
Customer Buy Call (std)	14.85** (6.658)	-2.337 (3.623)	1.947 (4.317)	1.859 (2.655)	14.35 (13.15)	13.11 (16.80)
Customer Sell Call (std)	-3.933 (4.218)	-9.437 (5.802)	-1.873 (4.682)	-0.4905 (1.913)	-21.89 (23.52)	13.61 (13.32)
Customer Buy Put (std)	-4.319 (5.805)	-10.75 (9.048)	-5.901 (4.199)	0.1897 (3.136)	12.09 (9.657)	11.74 (25.12)
Customer Sell Put (std)	1.165 (1.507)	0.0991 (2.743)	1.618 (3.355)	2.159 (4.062)	-5.944 (40.02)	-35.29 (53.62)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
F-test (1st stage), Buy Call	53,875.7	53,074.4	100,335.4	85,117.9	7,845.9	8,974.5
F-test (1st stage), Sell Call	67,045.3	65,774.5	112,009.4	100,761.6	4,686.3	3,860.0
F-test (1st stage), Buy Put	46,889.4	46,221.5	52,766.2	46,785.3	6,427.7	4,614.7
F-test (1st stage), Sell Put	167,744.5	92,938.0	41,352.4	32,768.1	825.13	781.90
Observations	7,337,832	7,015,567	4,384,249	4,180,466	833,295	795,659
R ²	0.00608	0.00202	0.02224	0.00325	0.02616	0.00721

The table reports coefficient estimates for fixed effects model of equation 13 estimated at a 10-minute frequency. All dependent variables are underlying excess returns (in percentage terms), annualized from 10-minute intervals. Concurrent returns correspond to the same 10-minute interval as options trading, while future returns measure subsequent stock performance. All four signed customer volumes are fitted values from equation 12 using professional customer volumes as instruments in the first stage. Control variables include lagged excess returns (20-minute and 10-minute previous realized returns to control for momentum effects), realized volatility, market volatility, rebalancing cost, and gamma exposure. All trader volumes are standardized. Two-way fixed effects include option-day and time fixed effects. Moneyness classification is based on stock-to-strike price ratio with $\pm 5\%$ threshold. Standard errors are two-way clustered by option-day and time and shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A3
Information Content of Customer Trading by Moneyness: 2SLS on MMs

Dependent Variables:	ATM		OTM		ITM	
	Concurrent (%) (10min) (1)	Future (%) (10min forward) (2)	Concurrent (%) (10min) (3)	Future (%) (10min forward) (4)	Concurrent (%) (10min) (5)	Future (%) (10min forward) (6)
Model:						
Customer Buy Call (std)	88.36*** (18.11)	1.546 (4.735)	37.00*** (12.33)	-0.3819 (3.147)	14.03 (26.22)	28.37 (44.88)
Customer Sell Call (std)	8.378 (7.024)	-1.629 (2.731)	4.369 (5.731)	-1.079 (4.123)	-1.078 (27.65)	-21.88 (20.61)
Customer Buy Put (std)	-28.41*** (9.283)	6.050 (5.053)	-15.84*** (6.029)	-2.086 (6.412)	-196.7 (188.3)	233.1 (283.7)
Customer Sell Put (std)	-11.04** (5.450)	0.5934 (3.707)	11.45** (5.795)	-5.155 (5.210)	-15.49 (15.99)	-9.508 (16.99)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
F-test (1st stage), Buy Call	622,897.9	558,272.0	396,839.9	379,211.4	482,778.2	491,196.6
F-test (1st stage), Sell Call	323,098.4	288,817.8	291,115.6	268,582.2	124,167.4	118,371.4
F-test (1st stage), Buy Put	163,576.6	149,771.7	603,232.3	589,156.9	8,938.0	8,105.1
F-test (1st stage), Sell Put	288,654.0	294,917.4	426,863.5	411,536.6	316,649.4	291,881.3
Observations	7,337,832	7,015,567	4,384,249	4,180,466	833,295	795,659
R ²	0.00614	0.00203	0.02222	0.00325	0.02646	0.00795

The table reports coefficient estimates from a fixed effects model of equation 13 with 10-minute frequency. All dependent variables are underlying excess returns (in percentage terms), annualized from 10-minute intervals. Concurrent returns correspond to the same 10-minute interval as the options trading, while future returns measure subsequent stock performance. All four signed customer volumes are fitted values from equation 12, using market maker volumes as instruments in the first stage. Control variables include lagged excess returns (20-minute and 10-minute prior realized returns to capture momentum effects), realized volatility, market volatility, rebalancing costs, and gamma exposure. All trader volumes are standardized. Two-way fixed effects include option-day and time fixed effects. Moneyness is classified based on the stock-to-strike price ratio within a $\pm 5\%$ threshold. Standard errors are two-way clustered by option-day and time and shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A4
Price Impact of Customer Trading by Moneyness: 2SLS Results

Dependent Variable:	Absolute Return (%) (10min concurrent)					
Moneyness:	ATM		OTM		ITM	
Model:	PCs (1)	MMs (2)	PCs (3)	MMs (4)	PCs (5)	MMs (6)
Customer Buy Call (std)	9.919*** (2.957)	50.46*** (11.57)	-1.840 (2.018)	25.04*** (7.522)	4.471 (7.121)	60.42* (33.21)
Customer Sell Call (std)	4.387 (3.577)	39.46*** (9.402)	0.4306 (1.563)	16.08* (8.534)	-24.75** (10.11)	93.02*** (31.66)
Customer Buy Put (std)	-1.517 (2.963)	25.87*** (5.932)	-4.516 (3.234)	27.91*** (6.965)	18.50 (26.08)	205.3 (196.9)
Customer Sell Put (std)	-3.758 (2.988)	26.56*** (8.115)	-3.250** (1.549)	18.20*** (4.911)	-42.28** (17.75)	11.20 (19.81)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
F-test (1st stage), Buy Call	54,382.0	694,964.3	106,288.7	453,524.2	7,548.7	564,715.7
F-test (1st stage), Sell Call	70,566.2	371,917.7	105,899.7	337,045.4	5,117.5	169,322.9
F-test (1st stage), Buy Put	40,917.8	166,390.2	52,152.4	648,675.1	6,404.2	13,043.8
F-test (1st stage), Sell Put	167,391.5	320,205.8	41,716.1	460,805.7	766.97	398,074.4
Observations	7,587,416	7,587,416	4,535,125	4,535,125	871,728	871,728
R ²	0.30660	0.30657	0.24529	0.24526	0.25333	0.25340

The table reports coefficient estimates for the price impact of customer options trading through two types of counterparties. Two-way fixed effects include option-day and time fixed effects. The dependent variable is the absolute concurrent 10-minute stock return (in percentage terms), annualized. Models 1–3 present 2SLS results in which all four signed customer volumes are instrumented with professional customer volumes, while Models 4–6 use market maker volumes as instruments. Control variables include underlying realized volatility, market volatility, and absolute gamma exposure. All trader volumes are standardized. Moneyness is classified based on the stock-to-strike price ratio within a $\pm 5\%$ threshold. Standard errors are two-way clustered by option-day and time and shown in parentheses (** p<0.01, * p<0.05, * p<0.1).

Table A5
Logistic Regression on Professional Trading Occurrence (Signed Volume)

Dependent Variable:	Professional Trade Indicator				Professional Trade Indicator				Professional Trade Indicator				Professional Trade Indicator			
Moneyneess:	Buy Call	Sell Call	Buy Put	Sell Put	Buy Call	Sell Call	Buy Put	Sell Put	Buy Call	Sell Call	Buy Put	Sell Put	Buy Call	Sell Call	Buy Put	Sell Put
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
Implied Volatility (std.)	-0.7254*** (0.0709)	-0.3087*** (0.0442)	0.0180 (0.0421)	0.1500*** (0.0317)	-0.5893*** (0.0602)	-0.1703*** (0.0445)	-0.0875* (0.0521)	0.0892* (0.0466)	-0.5371*** (0.1099)	-0.3996*** (0.0949)	0.1884*** (0.0458)	0.2689*** (0.0452)	-0.0230 (0.0876)	0.1296*** (0.0423)	0.2402*** (0.0855)	0.1839*** (0.0532)
Historic Volatility (std.)	0.1159*** (0.0199)	0.1824*** (0.0229)	0.0236 (0.0210)	0.0739*** (0.0175)	0.1844*** (0.0221)	0.2672*** (0.0226)	0.0775*** (0.0224)	0.1764*** (0.0222)	-0.1657*** (0.0384)	0.0602* (0.0326)	0.0627*** (0.0242)	0.0792*** (0.0183)	-0.0916* (0.0548)	-0.1010 (0.0673)	-0.3489*** (0.0750)	-0.1291* (0.0688)
EAD (Day Before)	-0.5367*** (0.0732)	-0.5334*** (0.0700)	-0.8465*** (0.0848)	-0.4797*** (0.0745)	-0.5681*** (0.0788)	-0.4960*** (0.0693)	-0.7186*** (0.0960)	-0.5121*** (0.0924)	-0.5758*** (0.1013)	-0.7730*** (0.1174)	-1.153*** (0.1183)	-0.5107*** (0.1066)	0.2003 (0.1531)	0.2114 (0.1941)	-0.0918 (0.2255)	0.2600 (0.2012)
EAD	-1.068*** (0.0993)	-0.3785*** (0.0668)	-0.9033*** (0.0798)	-0.1530** (0.0774)	-1.081*** (0.1006)	-0.1772*** (0.0671)	-0.7231*** (0.0907)	-0.0184 (0.0849)	-1.001*** (0.1429)	-0.8718*** (0.1212)	-1.164*** (0.1140)	-0.3363*** (0.1171)	-0.8020*** (0.2136)	-0.3735* (0.2099)	-1.468*** (0.2796)	-0.3994* (0.2097)
EAD (Day After)	-0.2903*** (0.0723)	0.1597*** (0.0461)	-0.2101*** (0.0620)	0.2136*** (0.0575)	-0.2840*** (0.0883)	0.2423*** (0.0542)	-0.2319*** (0.0758)	0.3009*** (0.0602)	-0.3161*** (0.0834)	-0.0133 (0.0666)	-0.1261 (0.0789)	0.0394 (0.1189)	-0.0762 (0.1711)	0.0291 (0.1953)	-0.3002 (0.2861)	-0.4319** (0.2189)
Medium Order (>100 contracts)	0.8081*** (0.0781)	0.8444*** (0.0631)	0.3349*** (0.1107)	0.2717* (0.1477)	0.7802*** (0.0797)	0.8202*** (0.0698)	0.2888** (0.1400)	0.2039 (0.1565)	0.6895*** (0.0626)	0.7440*** (0.0663)	0.3505*** (0.0756)	0.2434** (0.1048)	0.3860* (0.2332)	0.8214*** (0.2199)	-1.090** (0.5081)	-0.2262 (0.3902)
Large Order (>200 contracts)	1.342*** (0.0889)	1.354*** (0.1282)	1.194*** (0.0877)	1.265*** (0.1116)	1.227*** (0.0832)	1.106*** (0.1071)	1.023*** (0.1045)	0.9678*** (0.1311)	1.467*** (0.1293)	1.655*** (0.1923)	1.415*** (0.1202)	1.684*** (0.1601)	2.073*** (0.2233)	1.910*** (0.2668)	1.568*** (0.3395)	1.699*** (0.3128)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	13,091,914	13,088,891	13,093,694	13,081,394	7,636,792	7,635,148	7,637,900	7,622,243	4,581,119	4,549,077	4,577,413	4,569,668	775,087	779,449	777,466	743,498

The dependent variable is a dummy indicating specific professional trading activity (Buy Call, Sell Call, Buy Put, or Sell Put) within a 10-minute interval. Historic volatility is measured as stock volatility from the preceding 10-minute bucket, annualized and standardized. Implied volatility is computed using the Black–Scholes model and standardized. Control variables include delta, vega, rebalancing cost, relative stock and option bid–ask spreads, and days to expiration. Medium Order equals 1 if any single customer order exceeds 100 contracts within the 10-minute interval, while Large Order equals 1 if any single order exceeds 200 contracts. Earnings Announcement (EAD) is a dummy for the day of the announcement. Two-way fixed effects include option-day and time. Moneyneess is classified based on the stock-to-strike price ratio within a $\pm 5\%$ threshold. Standard errors are two-way clustered by option-day and time, shown in parentheses (** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$).

Table A6
Poisson Regression on Professional Trading Activity

Variable	Buy Call Trades				Sell Call Trades				Buy Put Trades				Sell Put Trades			
	Full Sample	ATM	OTM	ITM	Full Sample	ATM	OTM	ITM	Full Sample	ATM	OTM	ITM	Full Sample	ATM	OTM	ITM
Imp. Vol. (std.)	-0.6034*** (0.2105)	-0.6897** (0.2875)	-0.6137** (0.3031)	0.2132 (0.2924)	-0.1406 (0.1264)	-0.3041 (0.1997)	-0.2358 (0.2325)	0.1413 (0.2094)	0.3257 (0.2162)	0.4057** (0.1983)	0.0097 (0.1534)	0.0590 (0.1658)	-0.3069** (0.1216)	-0.1941 (0.1596)	-0.3625 (0.2455)	-0.1633* (0.0968)
Historic Volatility (std.)	0.0766** (0.0389)	0.1657*** (0.0239)	-0.1196 (0.1739)	-0.1640 (0.1110)	0.2126** (0.0890)	0.0515 (0.1322)	0.2291* (0.1310)	0.2255* (0.1335)	-0.1037 (0.0948)	-0.1336 (0.1633)	0.0223 (0.0662)	-0.1324 (0.1158)	-0.0994 (0.1198)	-0.0948 (0.1470)	-0.1236 (0.2084)	0.0372 (0.0377)
EAD (Before)	-0.1955 (0.4541)	0.1729 (0.5642)	-1.027*** (0.3470)	-0.6265 (0.8962)	0.0250 (0.3448)	0.3794 (0.3732)	0.3290 (0.3967)	-1.202** (0.5184)	-0.2542 (0.5692)	0.0529 (0.6905)	-0.9021* (0.4968)	-0.9506 (6.218)	0.6206 (0.5107)	0.2871 (0.4063)	0.5666 (0.6345)	-1.377** (0.6203)
EAD (Day)	-0.1445 (0.4759)	-0.7820** (0.3748)	0.1013 (0.7147)	0.6950 (0.5412)	-0.6297** (0.2969)	-0.7883** (0.3270)	-0.8792** (0.3802)	-0.1409 (0.6501)	-0.1062 (0.5219)	-0.2878 (0.6283)	0.1134 (0.5041)	0.1506 (0.6677)	-0.4002 (0.3881)	-0.3755 (0.3516)	-0.3431 (0.7482)	-1.392*** (0.4698)
EAD (After)	0.3377 (0.4579)	-0.3430 (0.3925)	0.8155 (0.8310)	0.1933 (0.6274)	-0.3108 (0.3692)	-0.3622 (0.3077)	-0.1944 (0.5735)	0.0227 (0.7437)	-0.2699 (0.2430)	-0.2493 (0.3563)	-0.3545 (0.3591)	-0.8374* (0.4294)	-0.1883 (0.4635)	0.0808 (0.6422)	-0.3329 (0.4783)	-0.0315 (0.5964)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	13,091,914	7,636,792	4,581,119	775,087	13,088,891	7,635,148	4,549,077	779,449	13,093,694	7,637,900	4,577,413	777,466	13,081,394	7,622,243	4,569,668	743,498

Panel shows Poisson regression results for the number of professional buy call, sell call, buy put, or sell put trades in a 10-minute interval, weighted by predicted probabilities from a logistic regression model of professional trade occurrence. Historic volatility is stock volatility from the past 10-minute bucket, annualized and standardized. Implied volatility is based on Black-Scholes and standardized. Controls include medium (a dummy equal to 1 if any single customer order exceeds 100 contracts within the 10-minute interval), large (a dummy equal to 1 if any single order exceeds 200 contracts within the 10-minute interval), delta, vega, rebalancing cost, relative stock and option spreads, market volatility, and days to expiration. Earnings Announcement (EAD) is the day of announcement; EAD (Before) is the day before; EAD (After) is the day after. Two-way fixed effects include option-day and time. Moneyness is based on stock-to-strike price ratio with $\pm 5\%$ threshold. Standard errors are two-way clustered by option-day and time, shown in parentheses (** $p < 0.01$, * $p < 0.05$, $p < 0.1$).

Table A7
Poisson Regression on Customer Trading Activity

Variable	Customer Buy Call Trades				Customer Sell Call Trades				Customer Buy Put Trades				Customer Sell Put Trades			
	Full Sample	ATM	OTM	ITM	Full Sample	ATM	OTM	ITM	Full Sample	ATM	OTM	ITM	Full Sample	ATM	OTM	ITM
Implied Volatility (std.)	-0.0066 (0.0191)	-0.0877*** (0.0161)	-0.1155*** (0.0388)	0.0853*** (0.0128)	-0.0851*** (0.0237)	0.0052 (0.0210)	-0.1147*** (0.0427)	-0.0647** (0.0276)	0.2169*** (0.0138)	0.2186*** (0.0162)	0.1970*** (0.0217)	0.1335*** (0.0250)	0.1776*** (0.0153)	0.1531*** (0.0244)	0.2245*** (0.0219)	-0.1205*** (0.0253)
Historic Volatility (std.)	0.0855*** (0.0114)	0.1809*** (0.0158)	0.0776*** (0.0124)	0.0473*** (0.0157)	0.0867*** (0.0112)	0.1704*** (0.0158)	0.0221 (0.0167)	0.0819*** (0.0158)	0.0581*** (0.0061)	0.0951*** (0.0207)	0.0986*** (0.0115)	0.0386*** (0.0034)	0.0443*** (0.0101)	0.0823*** (0.0148)	0.0935*** (0.0085)	-0.0110 (0.0312)
EAD (Before)	0.1038*** (0.0270)	0.0786** (0.0332)	0.1502*** (0.0479)	-0.0727 (0.0710)	0.0583* (0.0343)	-0.0140 (0.0398)	0.1526*** (0.0495)	0.0045 (0.0967)	0.0303 (0.0424)	-0.0582 (0.0577)	0.0539 (0.0533)	-0.1793 (0.1143)	0.0770* (0.0432)	0.0458 (0.0753)	0.0235 (0.0440)	-0.3449*** (0.1091)
EAD (Day)	0.0546*** (0.0190)	0.0827** (0.0400)	-0.0056 (0.0472)	0.0740 (0.0515)	0.0490 (0.0312)	0.0687* (0.0405)	-0.0155 (0.0458)	0.0295 (0.0657)	0.1223*** (0.0261)	0.1196*** (0.0275)	0.2154*** (0.0499)	-0.0686 (0.1356)	0.1316*** (0.0303)	0.1679*** (0.0421)	0.1607*** (0.0406)	0.1701 (0.1072)
EAD (After)	-0.0000 (0.0199)	-0.0080 (0.0198)	-0.0042 (0.0351)	-0.0010 (0.0602)	0.0097 (0.0264)	0.0218 (0.0251)	-0.0101 (0.0654)	-0.1292* (0.0665)	0.0717** (0.0315)	0.1070*** (0.0413)	0.0531 (0.0554)	-0.1258 (0.2069)	0.1339*** (0.0348)	0.1824*** (0.0439)	0.0955* (0.0532)	0.1886 (0.1649)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	13,099,558	7,648,887	4,599,466	851,205	13,099,558	7,648,887	4,599,466	851,205	13,099,558	7,648,887	4,599,466	851,205	13,099,558	7,648,887	4,599,466	851,205

The panel reports Poisson regression results for the number of customer Buy Call, Sell Call, Buy Put, or Sell Put trades within a 10-minute interval. Historic volatility is measured as stock volatility from the preceding 10-minute bucket, annualized and standardized. Implied volatility is computed using the Black-Scholes model and standardized. Control variables include medium (a dummy equal to 1 if any single customer order exceeds 100 contracts within the 10-minute interval), large (a dummy equal to 1 if any single order exceeds 200 contracts within the 10-minute interval), delta, vega, rebalancing cost, relative stock and option bid-ask spreads, market volatility, and days to expiration. Earnings Announcement (EAD) denotes the announcement day; EAD (Before) the day prior; and EAD (After) the following day. Two-way fixed effects include option-day and time. Moneyness is classified based on the stock-to-strike price ratio within a $\pm 5\%$ threshold. Standard errors are two-way clustered by option-day and time, shown in parentheses (***) $p < 0.01$, (**) $p < 0.05$, (*) $p < 0.1$).

Table A8
Effective Spreads

Dependent Variable:	OLS		Two-Stage	
	(1)	Effective Spread (%) (2)	(3)	(4)
Market Maker Volume (std)	-0.1077 (0.0760)	-0.0994 (0.0738)	-0.7268*** (0.1078)	-0.4929*** (0.1189)
Professional Volume (std)	0.0295*** (0.0072)		0.4681*** (0.1294)	
MM Volume (std) $\times$ Small Prof		-0.3081*** (0.0534)		0.0050 (0.0646)
MM Volume (std) $\times$ Medium Prof		0.1168 (0.0720)		0.1083 (0.1161)
MM Volume (std) $\times$ Large Prof		0.2719*** (0.0720)		0.4963*** (0.0965)
Controls	Yes	Yes	Yes	Yes
Fixed effects	Yes	Yes	Yes	Yes
F-test (1st stage), MM Volume			315,989.8	278,511.9
F-test (1st stage), Prof Volume			208,589.3	
Observations	13,099,559	13,099,559	13,099,559	13,099,559
R ²	0.02542	0.02542	0.02519	0.02536

All the volumes variables are standardized. Control variables include days to expiration, inverse option mid-quote, historic volatility, market volatility, implied volatility, stock bid-ask spread, absolute delta, rebalancing cost, and vega. *Small Prof* indicates any professional volume > 0 , *Medium Prof* indicates volume > 100 contracts, *Large Prof* indicates volume > 200 contracts. Models (1) and (2) are OLS regressions. Models (3) and (4) are two-stage regressions where total volume of market maker and professional customers are fitted by customer volumes from equation 14. The fitted values then used in the main regression. Two-way fixed effects include option-day and time fixed effects and standard errors are also two-way clustered and shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.